

Incident reporting and investigation under the AI act: some insights from aviation

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Abstract

This article examines the provisions of the Artificial Intelligence Act (AIA) on incident reporting and investigation and explores how the regulatory context of the AIA could benefit from the experience of aviation. Accounting for the differences between the AI and the aviation industries, it is concluded that the establishment of a strong safety culture is a key element, while a series of interventions could significantly improve the current AIA system on incident reporting: (i) clarification of the reporting duties, including the development of a common taxonomy of risk or harm, alongside the expansion of the duty's personal scope; (ii) disconnection of incident reporting and investigation from liability; (iii) establishment of voluntary reporting systems; (iv) provisions on confidentiality of the reports and protection of the reporters; (v) mandatory investigation of at least some serious incidents by an independent authority; and (vi) effective dissemination of safety information. A major enabler of all these is the development of a strong safety culture.

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INTRODUCTION

The EU Artificial Intelligence Act (AIA)¹ represents a landmark effort to regulate artificial intelligence (AI) systems within the EU market. Its objectives are to facilitate the free movement of AI products and services within the EU, support innovation, promote the uptake of human-centric and trustworthy AI, and protect the health, safety, and fundamental rights of EU citizens and residents.²

The AIA adopts a risk-based approach, establishing three categories of regulated systems. First, there are AI systems that pose unacceptable risk, and its use is prohibited.³ Second, there are high-risk AI systems, for which the AIA establishes extensive compliance requirements.⁴ Third, there are AI systems with limited risk to which transparency requirements apply.⁵ In addition, there are special rules on general-purpose AI models.⁶

The AIA applies to all legal entities or persons that are based in the EU, reside in the EU, or offer products and services in the EU, even if they are incorporated in a third (non-EU) State.⁷ The scope of the AIA covers products and services offered for civilian use, including dual use but not for purely scientific research and development.

To ensure safety monitoring and improvement of high-risk AI systems and of general-purpose AI models with systemic risk, the AIA's incident reporting framework mandates the prompt disclosure and investigation of serious incidents. This article examines the pertinent provisions of the AI Act and proposes improvements and supplementary measures based on the experience of the aviation industry. However, general, comprehensive recommendations for the improvement of the AIA incident reporting system lie outside of the scope of this article. Part I assesses incident reporting and investigation mechanisms under the AIA. Part II outlines key aspects of incident reporting and investigation in aviation, which has a remarkable safety record. Part III explores how the AIA can draw on aviation's experience to enhance its own approach, taking into account the differences between these industries. The article concludes with a summary of its findings.

INCIDENT REPORTING UNDER THE AI ACT

Article 73 AIA establishes a duty of developers, and in some cases deployers, of high-risk AI systems for reporting of ‘serious incidents’, investigation of the serious incident and taking remedial action while cooperating with the market surveillance authorities.

Similar reporting duties establishes Article 55(1)(c) for providers of general-purpose AI models with systemic risk. Guidance in the implementation of this provision can be derived from the voluntary Code of Practice for General-Purpose AI Models,⁸ prepared by independent experts and (at the time of this writing) awaiting approval by the Commission and the Member States. Once approved, the Code of Practice may be used for demonstrating compliance with Articles 53–55 AIA.⁹

It has been observed¹⁰ that the provisions of the AIA on serious incident reporting have been influenced by respective provisions of the Medical Devices Regulation¹¹ and the General Data Protection Regulation (GDPR).¹² The Commission also notes that the AIA incident monitoring aims to align with the OECD AI Incidents Monitor and Common Reporting Framework, where possible.¹³ Furthermore, Article 87 AIA refers to reports of infringements and protection of reporting persons under the Whistleblower Directive,¹⁴ which, under certain circumstances, may have the effect of voluntary reporting of incidents.

Duty to report

As part of their post-marketing duties,¹⁵ providers of high-risk AI systems, or deployers under certain circumstances, have to report any ‘serious incident’ to the market surveillance authorities of the Member States where that incident occurred.¹⁶ Moreover, they need to establish written policies, procedures, and instructions related to the reporting of serious incidents as part of their quality management system.¹⁷ The duty covers high-risk systems that have been placed on the market¹⁸ or are tested under real conditions outside regulatory sandboxes.¹⁹ Furthermore, there are special provisions on reporting duties of the

providers of general-purpose AI models with systemic risk,²⁰ which are specified in greater detail in the Code of Practice for GP AI models.

Objective

The purpose of post-marketing obligations, including reporting of serious incidents, is to ensure that providers of high-risk AI systems can take into account real-world use experience to improve system performance, improve design and development processes, and proceed to timely corrective action where necessary.²¹ These duties also facilitate the detection of emerging risks, particularly in AI systems that continue to ‘learn’ after deployment.²²

The draft guidelines of the Commission clarify that the reporting obligations have a threefold aim to: (i) create an early warning system of AI-system risks, (ii) establish accountability for providers and users, and (iii) enable timely corrective action by market surveillance authorities.²³ Thus, the provisions have a broader preventive effect that extends beyond any single AI system.²⁴

Persons under duty

The persons under duty to report serious incidents are providers and deployers of high-risk AI systems and providers of AI general-purpose models with systemic risk.

Providers and deployers of high-risk AI systems

The AIA provisions of serious incident reporting focus on the reporting duties of providers of high-risk AI systems.²⁵ A ‘provider’ is a natural person or legal entity, including public authorities, that develops or has developed an AI system and places it on the market, or puts the AI system into service under its own name or trademark, whether for payment or free of charge.²⁶

Furthermore, deployers of high-risk AI systems, ie professional users,²⁷ may have to submit a serious incident report under certain circumstances. In principle, deployers have to merely inform the provider, the importer or the distributor, and the market surveillance

authority of the incident.²⁸ Exceptionally, deployers have to submit a serious incident report when they are unable to inform the provider that a serious incident occurred.²⁹ This duty includes an interim report to ensure timely submission.³⁰

AI systems are classified as high-risk, according to the criteria set in Article 6 AIA. There are two categories of high-risk systems.

First, Article 6(1) provides that an AI system is considered to be high-risk, where both of the following conditions are fulfilled: (i) the AI system is a product or a safety component of a product, covered by specific EU legislation on product safety listed in Annex I, eg on toys, radio equipment, medical devices, etc.; (ii) pursuant to such EU legislation, a third-party (ie a party other than the developer of the AI system or AI model) must assess the conformity of that product with the applicable product-safety rules, as a condition for such product to be placed in the EU internal market.

Second, AI systems referred to in Annex III AIA are considered to be high-risk. These are AI systems used in (i) biometrics, (ii) critical infrastructure, (iii) education and vocational training, (iv) employment, workers' management and access to self-employment, (v) access to and enjoyment of essential private services, and essential public services and benefits, (vi) law enforcement, (vii) migration, asylum, and border control management, and (viii) administration of justice and democratic processes.

Providers of AI general-purpose models with systemic risk

In addition, providers of AI general-purpose models with systemic risks are under a duty to report serious incidents.³¹ The definitional elements of a provider of a general-purpose model are the same as these of a provider of a high-risk AI system.³²

The AIA defines a 'general-purpose AI model' as an AI model that displays significant generality and is capable of competently performing a wide range of distinct tasks, and that can be integrated into a variety of downstream systems or applications. The definition excludes AI models used for research, development, or prototyping activities before they are placed on the market. A general-purpose AI

model is distinct from a general-purpose AI system, which is an AI system based on a general-purpose AI model and able to serve various purposes, both for direct use and for integration in other AI systems.³³

A 'systemic risk' is defined as a risk that is specific to the high-impact capabilities of general-purpose AI models, ie capabilities that match or exceed the capabilities recorded in the most advanced general-purpose AI models,³⁴ when these models have a significant impact on the EU market due to their reach, or due to actual or reasonably foreseeable negative effects on public health, safety, public security, fundamental rights, or the society as a whole, and such risk can be propagated at scale across the value chain.³⁵

Under Article 51(1)(a) AIA, a general-purpose AI model is classified as a general-purpose AI model with systemic risk, if it has high-impact capabilities evaluated on the basis of appropriate technical tools and methodologies. Such capabilities are presumed, when the cumulative amount of computation used for its training exceeds 10^{25} floating point operations.³⁶ In the alternative, the European Commission may decide that an AI general-purpose model has high-impact capabilities, based on the criteria set out in Annex XIII AIA,³⁷ such as the number of parameters of the model, the quality or size of the data set, the amount of computation used for training the model, etc.

Serious incident

The AIA defines a '*serious incident*' as 'an incident or malfunctioning of an AI system that directly or indirectly leads to any of the following:

- a. the death of a person, or serious harm to a person's health;
- b. a serious and irreversible disruption of the management or operation of critical infrastructure;
- c. the infringement of obligations under Union law intended to protect fundamental rights;
- d. serious harm to property or the environment'.³⁸

The notion of 'serious incident' applies to the reporting duties of both AI system developers (or deployers in some cases) under Article 73 AIA

and developers of general-purpose AI models with systemic risk under Article 55(1)(c) AIA.³⁹ Moreover, in the case of the latter, the Code of Practice for GP AI models specifies that a ‘serious incident’ comprises a serious cybersecurity breach.⁴⁰

Incident

The AIA does not define the term ‘incident’.

The wording of the definition of ‘serious incident’ (*incident or malfunctioning*) seems to imply that an ‘incident’ is different than a ‘malfunction’, but this is not very accurate. The European Commission observes that there is no strict distinction between an ‘incident’ and a ‘malfunction’; the definition emphasises the importance of malfunctions in incident monitoring.⁴¹ The Commission further clarifies that a ‘malfunction’ is any function of the AI system contrary to its intended performance or to the performance stated by the provider in the technical documentation under Article 11 AIA.⁴²

Further guidance could provide the definition of the ‘AI incident’ developed by the Organization for Economic Cooperation and Development (OECD), to which the European Commission participated: ‘An AI incident is an event, circumstance or series of events where the development, use or malfunction of one or more AI systems directly or indirectly leads to any of the following harms’.⁴³ Thus, the term ‘incident’ encompasses situations related to the development or operation of a high-risk AI system or general-purpose AI model irrespective of whether they involve a malfunction. These situations must have caused specific harmful outcomes.

Serious

An incident is ‘serious’ if it is causally linked to specific, severe outcomes mentioned in points (a)–(d) of the definition. The wording of the definition (*led*) and the legislative history of the provision indicate that a specific outcome must have been produced,⁴⁴ so that ‘near-misses’ are not covered.⁴⁵

Specific outcomes

These outcomes are as follows:

- **Death of a person or serious harm to a person's health.**

The provision's generic wording entails that it covers both physical and mental impairments of health.⁴⁶ A serious harm may also be the deterioration of an already existing condition,⁴⁷ as implied also by the Commission guidance.⁴⁸

- **A serious and irreversible disruption of the management or operation of critical infrastructure.**

Critical infrastructure means an asset, a facility, equipment, a network or a system, or a part of an asset, a facility, equipment, a network or a system, which is necessary for the provision of an essential service.⁴⁹

The term 'essential service' means a service which is crucial for the maintenance of vital societal functions, economic activities, public health and safety, or the environment.⁵⁰ The incident must have caused a disruption of the management or operation, which is both serious and irreversible. A disruption is 'serious', when it might result (i) in an imminent threat to life or the physical safety of a person, including through serious harm to the provision of basic supplies to the population or to the exercise of the core function of the State, (ii) destruction of key infrastructure, and (iii) disruption in social and economic activities.⁵¹ The Commission seems to suggest that 'irreversibility' amounts to (near) total destruction of the critical infrastructure at hand.⁵² Such a view would excessively reduce the scope of the reporting duty. Therefore, irreversibility should be better understood as a disruption that requires extraordinary and extensive repairs to be restored.⁵³ The focus of the reporting duty should be on learning from experience, and such an objective is not promoted by narrowing its scope.

- **Serious harm to property or the environment.**

In view of the objective of Article 73 to promote information sharing for the improvement of safety, the term 'property' should be broadly understood as comprising both material and immaterial assets, such as intellectual property rights, e-money, and crypto assets. The seriousness of harm to property will depend on the value of the damage caused, the difficulty to restore the damage or its irreversibility, etc.⁵⁴

To define ‘serious’ harm to the environment, the Environmental Liability Directive (ELD)⁵⁵ and the Environmental Crime Directive (ECD)⁵⁶ could serve as guidance.⁵⁷ Article 2(1)(a) of the Environmental Liability Directive defines ‘environmental damage’ as damage to protected species and natural habitats, water, and land. Article 3(6) ECD provides that the assessment of whether damage is substantial includes factors such as (i) the baseline condition of the affected environment, (ii) whether the damage is long-lasting, medium-term or short-term; (iii) the extent of the damage, and (iv) the reversibility of the damage. The Commission mentions as further examples the contamination of environmental resources and the disruption of natural ecosystems.⁵⁸ To this, the creation of orbital debris could be added, which has adverse consequences for the use of outer space.⁵⁹

- **Infringement of obligations under Union law intended to protect fundamental rights.**

The term ‘fundamental rights’ refers to the EU Charter of Fundamental Rights.⁶⁰ The generic wording of this provision may include types of harm, such as violations of privacy, discrimination, violations of the right to a fair trial, violations of human dignity, mis- and disinformation, etc. The harmful outcome in this case seems to be the violation of legal provisions, not the actual harm that may result therefrom. Although not specifically mentioned, the violation needs to be ‘serious’, so that minor infringements of such obligations are not covered.⁶¹ The Commission clarifies that reportable are only those infringements that significantly interfere with Charter-protected rights on a large scale, citing as examples an AI-based recruitment system that excludes candidates based on ethnicity or gender; a credit scoring system that excludes certain categories of persons; a biometric identification system that frequently wrongly identifies people of different ethical background.⁶² These examples indicate that the provision is meant to apply mainly to cases of discrimination and privacy violations.

Nevertheless, a series of practical questions are raised.

First, it is unclear how the ‘infringement’ of such obligations should be established: would a formal determination be required, like a decision

of a supervisory authority, or would a mere potential infringement suffice to trigger the reporting duty? The latter appears more appropriate to avoid delays in reporting, given the need to preserve evidence and promptly investigate the serious incident. However, the threshold of likelihood of an infringement remains uncertain.

Second, there is a plethora of provisions under EU law aiming at protecting fundamental rights. In a sense, almost all legal provisions can be linked to the protection of one or more fundamental rights, while fundamental rights often need to be balanced against each other. This creates considerable legal uncertainty.

Especially for developers of general-purpose AI models with systemic risk, the Code of Practice for GP AI models refers to a ‘serious cybersecurity breach, including the (self-) exfiltration of model weights and cyberattacks’ as a distinct case of a serious incident. This is related to the duty of developers of general-purpose AI models with systemic risk to ensure an adequate level of cybersecurity protection for the model and its physical infrastructure throughout the model’s lifecycle.⁶³ In this regard, recital (115) AIA clarifies that cybersecurity risks include instances like accidental model leakage, unauthorised releases, circumvention of safety measures and defences against cyberattacks, unauthorised access, and model theft, which entail that developers are expected to secure model weights, algorithms, servers, and data sets. As a result, any cybersecurity breach that significantly affects such elements, for example, data poisoning or prompt injection,⁶⁴ would qualify as a ‘serious cybersecurity breach’.

In sum, the definition of ‘serious incident’ covers a wide variety of cases. However, the definition is not sufficiently clear, especially regarding the incidents involving violations of rules intended to protect human rights. Moreover, in some cases, it does not appear sufficiently coherent, when eg reportable is serious damage to property, but for critical infrastructure a nearly total destruction may be required. These shortcomings entail compliance challenges for developers and deployers, as well as enforcement issues for market surveillance authorities.

Causal link

There must be a direct or indirect causal link between the incident and the serious harm mentioned above, which means that, on the balance of probabilities, but for the incident, the serious harm would not have occurred.⁶⁵

An indirect causal link exists when the harmful outcome is a secondary effect of the incident, eg an AI-based credit scoring system incorrectly indicates that a person is not solvent enough and credit is denied; an AI system classifies a patient as low risk, thus preventing early detection of a deteriorating condition.⁶⁶

The Commission observes that causation does not exist in cases of unexpected misuse of the AI system.⁶⁷ However, such a restriction only makes sense in a liability context, to avoid overextension of liability. In incident reporting, the focus should be on learning from experience and preventing future harm. In this regard, reporting harms due to unforeseeable misuse provides valuable safety insights. Although an event could be seen as unforeseeable the first time it is reported, this may not be valid in case of subsequent similar events, eg chats with AI systems leading to suicides.⁶⁸

Deadlines for submission

Issues related to the deadlines for reporting a serious incident regard the length of the deadline and its initiating time point.

AI system providers and deployers

Article 73 establishes short deadlines for reporting, reflecting the increased risks anticipated by high-risk AI systems.⁶⁹

As a rule, the serious incident report must be made immediately after the provider has established a causal link between the AI system and the serious incident, or the reasonable likelihood of such a link.⁷⁰ The reference to the reasonable likelihood of a causal link considers potential challenges in connecting the harm that occurred with the (mal)function of the AI system.⁷¹ To avoid excessive delays in the reporting, the incident has to be reported not later than 15 days after

the provider or the deployer becomes aware of the serious incident, irrespective of causation issues.⁷² In this regard, 15 days is the maximum period allowed, which can be shortened in more severe cases.⁷³ The severity relates to the extent of the harm, the increased possibility of imminent further harm, the reversibility of harm, etc.⁷⁴ The exact length of the shortened period is case-dependent, and in many cases, it will be up to the market surveillance authority to determine retroactively whether there was an unjustifiable delay in reporting, although the report had been submitted within 15 days.

For specific severe cases, shorter, exceptional deadlines are established in the interest of public safety. First, if a person dies, the report must be provided immediately after the provider or the deployer has established or suspects a causal relationship between the high-risk AI system and the serious incident, but not later than 10 days after the date on which the provider or the deployer becomes aware of the serious incident.⁷⁵ Second, in the event of (i) a widespread infringement,⁷⁶ such as harm related to environmental protection, public health, or the functioning of democratic institutions,⁷⁷ or (ii) a serious and irreversible disruption of the management or operation of critical infrastructure, the report must be provided immediately, and not later than 2 days after the provider or the deployer becomes aware of that incident.⁷⁸

As a result, the more serious the harm occurred, the shorter are the deadlines for reporting. Interestingly, depending on the severity of the harm, the requirements of establishing a causal link to trigger the deadline also differ. In principle, a reasonable likelihood of causal connection is required,⁷⁹ yet in cases of death, a mere 'suspicion' suffices,⁸⁰ while in cases of widespread infringement or severe harm to critical infrastructure no causal link at all is required.⁸¹ Nonetheless, also in the third set of cases, at least an apparent correlation between the malfunction and the harm is necessary; otherwise, the provision would be practically impossible to implement.⁸²

To ensure timely submission, an initial, incomplete report can be filed, followed by a complete report.⁸³ This possibility enables appropriate management of complex incidents, creates legal certainty for AI system providers and deployers, and enables market surveillance authorities to make a first assessment of the case and order preliminary measures.⁸⁴

Providers of general-purpose AI models with systemic risks

Art. 55(1)(c) AIA obliges providers of AI general-purpose models with systemic risk to 'report, without undue delay, to the AI Office and, as appropriate, to national competent authorities'. The Code of Practice for GP AI models clarifies the applicable deadlines,⁸⁵ which coincide with the provisions applicable to providers and deployers of AI systems under Article 73. For the additional cases of serious cybersecurity breaches, the Code of Practice establishes a deadline of 5 days after the providers become aware of the involvement of their model in the incident.

The Code of Practice for GP AI models allows providers of AI general-purpose models with systemic risk to submit an interim report for unresolved incidents and update it every 4 weeks.⁸⁶ They have to submit a final report within 60 days after the incident has been resolved.⁸⁷ If multiple similar serious incidents occur within the reporting timelines, they may include them in the report(s) of the first serious incident while respecting the timelines for reporting for the first serious incident.⁸⁸

Furthermore, the Code of Practice lays down a duty of providers of AI general-purpose models with systemic risk to retain documentation of all relevant information gathered during the serious incident investigation for at least 5 years from the date of the documentation or the date of the serious incident, whichever is later, without prejudice to Union law applicable to such information.⁸⁹

Content of the report

Article 73 AIA does not clarify the content of the report to be submitted. However, some necessary elements can be deduced from the regulatory context of the provision (systematic interpretation).⁹⁰ The report should include all details necessary to identify the occurrence, such as the identification of the AI system; the harm that occurred; the facts of the case, including any (suspected) malfunctions of the system and their potential connection to the harm that occurred; identification of potential further risks; description of any remedial actions taken, planned and/or proposed.⁹¹ Especially for providers of general-purpose

models with systemic risk, the inclusion of potential corrective measures in the report is mandatory.⁹²

More detailed guidance can be derived from the Code of Practice for GP AI models. Measure 9.2 of the Code of Practice establishes a duty of providers of AI models with systemic risk to report at least the following information, redacted to the extent necessary to comply with other Union law applicable to such information:

1. the start and end dates of the serious incident, or best approximations thereof if the precise dates are unclear;
2. the resulting harm and the victim or affected group of the serious incident;
3. the chain of events that (directly or indirectly) led to the serious incident;
4. the model involved in the serious incident;
5. a description of material available setting out the model's involvement in the serious incident;
6. what, if anything, the provider intends to do or has done in response to the serious incident;
7. what, if anything, the provider recommends the AI Office and, as applicable, national competent authorities to do in response to the serious incident;
8. a root cause analysis with a description of the model's outputs that (directly or indirectly) led to the serious incident and the factors that contributed to their generation, including the inputs used and any failures or circumventions of systemic risk mitigations; and
9. any patterns detected during post-market monitoring that can reasonably be connected to the serious incident, such as individual or aggregate data on near misses.

Recipients of the report

AI system providers must submit the report to the market surveillance authorities of the Member States where the incident occurred.⁹³ The 'place of occurrence' is not defined. The wording of the provision (*States*) indicates that there can be more than one place of occurrence, namely the place where the operation or malfunction of the AI system took place,⁹⁴ as well as the place(s) where the damage occurred.⁹⁵

Providers of general-purpose AI models with systemic risk must report any serious incident to the AI Office, which forms part of the European Commission,⁹⁶ and, as appropriate, to national authorities.⁹⁷

From a compliance perspective, the fact that more than one State may be affected and each State may have designated more than one market surveillance authority⁹⁸ does not facilitate the reporting procedure. One issue is the identification of the competent authority for the particular incident in States other than the State where the provider or deployer is based. In such cases, it should suffice that the provider notifies the national single point of contact in these States that has been designated under Article 70(2) AIA. A similar issue is the identification of the competent authority for the specific incident in the 'home State', when that State has designated more than one authority. To facilitate compliance, it should be accepted that, if such authority is not obvious from the facts of the case (eg the incident concerns personal data leak apparently unrelated to cyber security and the personal data supervisory authority has been designated as competent), then the report can be submitted only to the national single point of contact.

Alleviated reporting requirements

The AIA establishes lighter reporting requirements for providers of certain AI systems in order to avoid duplication of reporting duties and facilitate compliance.⁹⁹

First, under Article 73(9) AIA providers of high-risk AI systems referred to in Annex III AIA¹⁰⁰ need to report only serious incidents regarding infringements of obligations under EU law intended to protect fundamental rights. There are two conditions for such alleviated duty: first, such systems are governed by other EU legal instruments; second,

these instruments establish equivalent reporting duties. The term 'equivalent' should be interpreted as referring to duties to report safety incidents to supervisory authorities, since the objective of the provision is the simplification of the reporting tasks.¹⁰¹

Similarly, Article 73 (10) provides that in cases of high-risk AI systems used as medical devices or as a safety component thereof, the providers have to report only serious incidents concerning infringements of obligations under EU law intended to protect fundamental rights. It suffices that the reports are submitted to the national competent authorities designated under the Medical Devices Regulation¹⁰² and the *In Vitro* Diagnostic Medical Devices Regulation¹⁰³ by the State where the incident occurred.¹⁰⁴

Since the AIA provides for alleviated reporting duties only regarding specific cases, reporting duties under other pieces of EU legislation remain unaffected (*expressio unius est exclusio alterius*). Nevertheless, Article 2(2) AIA entails that the reporting duty does not encompass AI systems covered by the Union harmonization legislation listed in Section B of Annex I.¹⁰⁵ These are AI systems used in the transportation sector, eg civil aviation, motor vehicles, railways, etc.

Follow up actions

Following the reporting, both providers and the supervisory authorities need to proceed to a series of actions.

AI system providers

The providers of the AI systems have to investigate the incident without delay.¹⁰⁶ The investigation includes a risk assessment of the incident and corrective action,¹⁰⁷ as part of the general risk-management duties of AI system developers under Article 9 AIA.¹⁰⁸ It requires the collection and analysis of all relevant facts and data to reconstruct the chain of events connecting the operation of the AI system and the harm that occurred.¹⁰⁹ However, in practice, establishing causal relationships may not be straightforward, since AI systems are very complex and opaque, while their output may be influenced by extrinsic factors, such as environmental conditions and user input.

Under Article 9(2) AIA, the risk management of AI systems is understood as a continuous iterative process planned and run throughout the entire lifecycle of the AI system, requiring regular systematic review and updating. Therefore, the occurrence of an incident implies the need to proceed anew to a risk assessment of the whole system, to identify wider or latent issues and determine the appropriate corrective measures.¹¹⁰ Such measures are also subject to a risk assessment since they may bring about new risks.¹¹¹

During the investigation, AI system providers have to cooperate with the competent authorities and, where relevant, with the notified body concerned.¹¹² If the investigation involves any altering of the AI system that may affect the subsequent evaluation of the causes of the incident, they must first inform the competent authorities of such action.¹¹³ This accounts for the need to preserve forensic evidence, which is balanced against the need to make system changes for remedial purposes.¹¹⁴

When assessing whether such an alteration has occurred, various factors should be considered: alterations to components directly involved in the serious incident, like software updates, hardware replacements, or configuration changes; changes affecting the availability of data required for technical investigations (such as overwriting training datasets or disabling monitoring tools) or influencing the ability to reconstruct the sequence of events leading to the incident (such as modification of log files, sensor data, or decision-making algorithms).¹¹⁵

These provisions entail that the AI system provider is primarily responsible for the investigation of the incident, since the provider knows best how the system was developed, trained, fine-tuned, etc, as well as its operational record so far. Nevertheless, the competent authorities must be kept informed on the progress of the investigation and be consulted on any actions that might undermine the efforts to establish or verify the causes of the incident.

If high-risk AI systems are being tested in real-world conditions outside AI regulatory sandboxes, the (prospective) providers have to adopt immediate mitigation measures or, failing that, suspend the testing in real-world conditions until such mitigation takes place, or otherwise terminate it.¹¹⁶ Upon termination of the testing, the (prospective)

providers have to establish a procedure for the prompt recall of the AI system.¹¹⁷

Although not explicitly mentioned, Article 73(6) entails that the AI system providers have to prepare and submit to the market surveillance authorities an incident investigation report. The incident investigation report is distinct from the reporting of the incident as such, although there may be some overlaps in their content. The reporting of the incident concerns the notification to the competent authorities of major parameters of the incident, so that these become aware of the incident and decide on any immediate mitigating or corrective measures. The incident investigation report is a thorough analysis of the incident and includes an established factual timeline with appropriate details, conclusions on the root causes and contributing factors, an analysis of the incident's impact on the safety of the AI system and other AI systems, the corrective measures taken, and any further recommendations from a technical and/or policy perspective.

Providers of general-purpose AI models with systemic risks

The duty of providers of general-purpose AI models with systemic risks to investigate the serious incident is implied by their generic duty to assess and mitigate systemic risks throughout the lifecycle of their model and to report serious incidents. According to the Code of Practice for GP AI models, providers have to investigate the causes and effects of serious incidents and the information related to the necessary elements of their incident notification report, in order to update the systemic risk assessment of their model.

The Code of Practice does not seem to distinguish between serious incident notification reports and investigation reports. Nevertheless, the investigation of the serious incident and the reassessment of the systemic risk cannot be completed without the submission of an investigation report to the AI Office and national competent authorities. The difference between a notification and an investigation report could consist of the level of detail. In this regard, the reference of the Code of Practice to a final serious incident report within 60 days after the serious incident has been 'resolved', which includes a root cause

analysis and recommendations, corresponds rather to an incident investigation report.

Market surveillance authorities

The market surveillance authorities need to decide, within 7 days from the reporting of the serious incident, on the appropriate measures to eliminate the risk posed.¹¹⁸ In extreme cases and as a last resort, they may order a recall or withdrawal from the market, or even the prohibition of the AI system.¹¹⁹ If the AI system is being tested in real-world conditions, then the surveillance authorities may (i) suspend or terminate the testing in real-world conditions or (ii) require the (prospective) provider or deployer to modify any aspect of the testing.¹²⁰

These provisions entail that the market surveillance authorities could proceed to a full investigation and anew evaluation of the AI system. Thus, the reporting is connected to potential administrative sanctions for regulatory non-compliance—notwithstanding any repercussions for the provider's reputation and competitiveness. At the same time, the provider is responsible for conducting the investigation, despite the harsh competition with other providers on the market. As a result, there are reduced incentives for AI system providers to report incidents and conduct a thorough investigation of their root causes. Although heavy fines for non-compliance may be imposed,¹²¹ the experience from the enforcement of other legal instruments indicates that tech companies sometimes perceive fines as the cost of doing business.¹²² In other words, the strong connection of reporting with supervision and enforcement is not very conducive to the improvement of the safety culture of AI system providers.

Dissemination of information

National market surveillance authorities need to notify the Commission of the serious incident, including any measures ordered. The notification occurs through the Safety Gate Rapid Alert System, a special EU central platform, developed and administered by the Commission, which serves the information exchange among the MS

and the Commission, and in which MS register information about serious product risks and corrective measures taken by market surveillance authorities and by economic operators.¹²³

Moreover, upon receiving a notification about a serious incident, market surveillance authorities have to inform the national public authorities or bodies that supervise or enforce duties under Union law protecting fundamental rights.¹²⁴ The latter are authorities identified by each MS and notified to the Commission.¹²⁵

In the case of AI regulatory sandboxes, national competent authorities have to submit yearly reports to the AI Office and to the European AI Board¹²⁶ on the progress and results of the implementation of the regulatory sandboxes, which include incidents and lessons learnt.¹²⁷

Reporting of breaches and protection of reporting persons

Article 87 AIA renders the Whistleblower Directive¹²⁸ applicable to the reporting of AIA infringements and the protection of persons reporting such infringements.

The Whistleblower Directive aims to protect persons working in the private or public sector who acquired information on breaches of EU law in a work-related context and wish to report them.¹²⁹ The Directive requires EU MS to ensure that there are effective channels to report breaches of EU law both internally, ie within the organization concerned, and externally.¹³⁰ There has to be an investigation and follow-up action by both the organization concerned and the EU MS,¹³¹ while whistleblowers are protected from retaliation.¹³² The identity of the reporting persons remains confidential, notwithstanding the protection of their personal data under the applicable Union instruments.¹³³ Organizations and MS have to keep records of every report received.¹³⁴

It should be noted that the reporting of ‘infringements’, or ‘breaches’ under the Whistleblower Directive, is different from the reporting of ‘serious incidents’ under the AIA. An ‘infringement’ under the Directive is an unlawful behaviour,¹³⁵ whereas a ‘serious incident’ under the AIA

is not necessarily connected to such behaviour. The two terms are distinct, although they could occasionally overlap. To the extent that an infringement constitutes also a serious incident, the reporting channels under the Whistleblower Directive will play the role of voluntary incident reporting systems.

Overall evaluation and international perspectives

The incident reporting system of the AI Act contains some very useful elements, such as reporting duties, specific deadlines, follow-up measures, a duty to investigate, and dissemination of information to the authorities of other MS and the AI Office. Hence, it recognizes the importance of collecting, analysing, and sharing safety information in a wider system of safety assurance of high-risk AI systems.

Nevertheless, its application largely depends on the providers and deployers of AI systems (or general-purpose models with systemic risks), while independent investigation appears to be a rather narrow exception. In addition, there are several unclear issues surrounding the notion of the 'serious incident', the content of the incident report, and the investigation procedure, which complicate both compliance and enforcement. The fact that near misses are not covered by the reporting duties reduces the scope of safety information coming into light, although it could be explained by an effort to reduce regulatory burden.

From an international perspective, the provisions of the AIA on incident reporting are a major step forward. Other jurisdictions do not provide at all for AI incident reporting across sectors, since they have not enacted horizontal AI safety legislation or provide only for generic reporting duties, such as China¹³⁶ and South Korea.¹³⁷ The absence of widely accepted international rules, even of a non-binding nature, complicates the practical implementation of the AIA for both third-country AI providers offering services in the EU and (prospective) EU AI providers intending to provide services in third countries. The fragmentation of regulatory regimes increases compliance costs, which may create tensions also at a political level.¹³⁸

As a result, the incident reporting provisions of the AIA are an important step to the right direction, yet there is considerable potential

for improvement. In this regard, the occurrence reporting and investigation system used in aviation might be able to assist.

OCCURRENCE REPORTING AND INVESTIGATION IN AVIATION

The aviation industry has a remarkable safety record.¹³⁹ One of the cornerstones of aviation safety is the international rules on reporting of safety occurrences and investigation of accidents. As it will be analysed below, occurrence reporting and accident investigation are based on some fundamental features: (i) mandating reporting systems both at the industry and at State level, with clear reporting requirements and a standardized risk classification; (ii) creating a non-punitive environment and ensuring confidentiality of the reported information, to foster a strong safety culture at all levels; (iii) accident investigation by an independent authority; (iv) obliging all actors to proceed to follow-up actions on the lessons learned; and (v) focusing on systems thinking and combining reactive with proactive safety measures.

Nonetheless, a preliminary issue is whether there are enough parallels between the aviation and the AI industry to justify the suitability of aviation as an example for the AI industry. Therefore, in this part, we will first examine the parallels between the aviation and the AI industry. After outlining the aviation regulatory system, we will explore the international rules on safety reporting and accident investigation, as implemented in the EU.

Parallels between aviation and the AI industry

There are several similarities between the aviation and the AI industry.

Complex systems

Both aviation and the AI industry are complex sociotechnical systems from a safety perspective. A complex system consists of many interacting aspects, parts, or functions, which together form a whole that has different qualities than its parts, and which cannot be easily

described, analysed, or predicted because of its intricacy and internal interactions.¹⁴⁰

In AI systems, especially the ones using machine learning, training data, model architectures, feedback systems, human oversight mechanisms, deployment environments, and other factors interact with each other in opaque ways. Things are further complicated by emergent behaviours, namely qualities and capabilities that had not been intended or anticipated by the system developer, which require special attention from a regulatory perspective.¹⁴¹ Aviation safety faces similar challenges, where complicated interactions occur among people in different organizations (flight crew, air traffic controllers, mechanics, flight dispatchers, etc.), working in different environments (aircraft, control towers, company offices, hangars, etc.), using sophisticated equipment with many subsystems and often automated functions.

A special complexity parameter is the human–machine interaction. In aviation, focusing too much on technical solutions, while overlooking important parameters related to human factors and human performance,¹⁴² has led to numerous safety incidents (near-misses) and some tragic accidents.¹⁴³ In the AI industry, although the exact effects of human–machine interaction remain largely understudied, serious incidents have occurred in a variety of contexts, in which significant contributing factors were related to human factors, such as overreliance on technology (automation bias) in accidents of automated vehicles,¹⁴⁴ or the tendency to see machines as humans under certain circumstances (anthropomorphism) in suicides.¹⁴⁵ As AI becomes more integrated with human decision-making and operational tasks, understanding and managing the human–AI interface is paramount for safety.

Consequently, in both industries, the need for a systems approach to safety has been highlighted. Safety failures and shortcomings are often unable to be explained by mere cause and effect constructions, while risks may cascade in unpredictable ways.¹⁴⁶ In aviation, traditional reactive methods, which focus on malfunctions and causal relationships (often called ‘Safety I’), are combined with proactive methods, which aim to understand how everyday activities may fail by understanding how such activities usually succeed (‘Safety II’).¹⁴⁷

Aviation considers it important to increase the resilience of the whole system by examining the complexity of the operating socio-technical parameters, anticipating errors and malfunctions, which are unavoidable and sometimes unpredictable, and having complementary safeguards in place, to avoid catastrophic failures.¹⁴⁸ Similarly, for the AI industry, the design of redundant paths and error tolerance on the entire sociotechnical system has been recommended.¹⁴⁹

Significant potential for harm

Both industries may cause significant harm, if safety is overlooked.

Aviation accidents can cause hundreds of fatalities and serious injuries at once, alongside enormous property damage. The economic impact of accidents may be multiplied by cascading effects, eg due to grounding of a whole series of aircraft.¹⁵⁰ Furthermore, serious safety occurrences can undermine public trust towards the entities involved.¹⁵¹

AI safety failures have even greater potential for large-scale harm, given that they are used across sectors. As a result, they can, among others, affect critical infrastructure, damage political discourse and undermine institutions, promote hate speech, damage the environment, increase inequalities, or even pose an existential risk to humankind, according to some estimates.¹⁵²

Data dependence and continuous learning

Safety improvements in aviation are heavily data-driven. The aviation industry collects vast amounts of safety-related data, from a range of sources, such as accident investigation reports, safety reporting systems, system data, and performance monitoring programmes, etc.¹⁵³ These data are used to continuously improve procedures, training, and aircraft design in a manner both reactive, ie as a response to specific safety occurrences, and proactive, ie by observing trends in performance that facilitate the anticipation of future occurrences.¹⁵⁴

AI systems and models are inherently data-based and data-dependent. Their overall performance, safety, and reliability are directly linked to

the quantity and quality of their training data and feedback mechanisms.¹⁵⁵

International operations and effect on national prestige and strategic competition

The operations of both industries have a very strong international element. Modern aviation is based on international operations across hundreds of countries. As a result, it is a major factor of economic growth at both international¹⁵⁶ and national levels.¹⁵⁷ In a similar way, AI applications are ubiquitous worldwide and its impact on the world economy has been characterized as transformational.¹⁵⁸

Despite their strong international character, both industries are linked to national prestige and strategic competition. National air carriers and airports serving international air traffic are seen as symbols of national economic power and are sometimes used as soft power instruments to increase the geopolitical standing of States.¹⁵⁹ AI technology connotes technological prowess and is increasingly seen as a major asset and catalyst in global strategic competition among States.¹⁶⁰ Consequently, in both industries, safety shortcomings, both technical and regulatory, can have wider economic and political effects.

Intermediate conclusion

The above show that there are significant parallels between the aviation and the AI industry. The complexity of both industries and their high stakes at multiple levels entail that safety assurance in both industries depends on collection, systematization, and analysis of safety information, and implementation of the lessons learned for constant improvement.

As aviation is probably the most dynamic industry in terms of safety performance,¹⁶¹ its regulations on safety reporting and investigation should be examined. To understand these regulations, first the overall regulatory system of aviation safety needs to be considered.

The aviation safety regulatory system in general

Aviation safety is regulated mainly at the international level by the International Civil Aviation Organization (ICAO). During the last years of World War II, studies initiated into the problems that civil aviation would face in the post-war period concluded that without an international regulatory framework aviation would not be a significant contributor in the post-war economic development.¹⁶²

Article 37 of the Convention on International Civil Aviation (Chicago Convention)¹⁶³ provides that each contracting State undertakes to collaborate in securing the highest practicable degree of uniformity in regulations, standards, procedures, and organization in relation to aircraft, personnel, airways, and auxiliary services, to facilitate and improve air navigation. To this end, ICAO adopts and amends international Standards And Recommended Practices (SARPS) dealing with various issues, including the investigation of accidents.¹⁶⁴ Article 38 Chicago Convention obliges States to immediately notify ICAO of any deviations of its own regulations and practices from SARPS. SARPS are designated as 'Annexes' to the Chicago Convention.¹⁶⁵ ICAO Member States undertake to incorporate SARPS in their national legislation. Moreover, ICAO provides implementing provisions and guidance on SARPS through a variety of technical documents, including the ICAO Manuals.

It is disputed whether the Annexes are an integral part of the Convention and binding on the Member States.¹⁶⁶ Nonetheless, the debate has little practical importance. States are strongly motivated to implement international standards because non-compliance with SARPs could eliminate the State concerned from any meaningful participation in international air navigation and air transport.¹⁶⁷

Within the EU legal order, the EU has shared competence with its MS on aviation.¹⁶⁸ The EU has issued legislative instruments on incident reporting and accident investigation to implement related ICAO Annexes. These will be examined below.

‘Occurrences’, ‘accidents,’ and ‘incidents’

ICAO Annexes refer to safety-related events as ‘occurrences’, which are distinguished into ‘accidents’ and ‘incidents’.

An ‘accident’ is an occurrence associated with the operation of an aircraft, in which (i) a person is fatally or seriously injured or (ii) the aircraft sustains damage or structural failure that needs to be repaired; or (iii) the aircraft is missing or is completely inaccessible.¹⁶⁹

An ‘incident’ is an occurrence, other than an accident, associated with the operation of an aircraft that affects or could affect the safety of operation.¹⁷⁰

A ‘serious incident’ is an incident involving circumstances indicating that there was a high probability of an accident and is associated with the operation of an aircraft.¹⁷¹ The difference between an ‘accident’ and a ‘serious incident’ lies only in the result.¹⁷²

EU law defines an occurrence as any safety-related event that endangers or that, if not corrected or addressed, could endanger an aircraft, its occupants, or any other person.¹⁷³

Safety reporting systems

Safety occurrences are registered in safety reporting systems, which are established and managed at different levels: at the level of individual aviation organizations, at the State level, and at the EU level.

Duty to establish safety reporting systems

States are under a duty to establish a comprehensive State Safety Management system, which includes mandatory and voluntary reporting systems, with the aim to register and analyse safety data and information.¹⁷⁴ States should establish processes for a comprehensive analysis of data and information from the safety databases to identify systemic and cross-cutting hazards.¹⁷⁵ To facilitate information sharing and exchange, the data should adhere to a standardized taxonomy.¹⁷⁶

Reporting systems should be user-friendly and designed to capture all valuable information concerning an occurrence, such as what happened

exactly, where and when.¹⁷⁷ Such an approach to safety management reflects the growing complexity of aviation as air traffic increases and socio-technical systems evolve.¹⁷⁸

There are mandatory and voluntary reporting systems. This reflects aviation's recognition that safety emerges not merely from regulatory compliance but from a culture of proactive disclosure.¹⁷⁹

Mandatory reporting systems

Mandatory reporting systems regard both accidents and incidents. In the EU, mandatory reporting applies to occurrences representing a significant risk to aviation safety and falling within a set of predefined categories, which relate to the main domains of the aviation activity.¹⁸⁰ The obligation to establish mandatory reporting systems lies with both Member States and aviation organizations, including aircraft operators, airport authorities, manufacturers, and maintenance providers.¹⁸¹ The European Union Aviation Safety Agency (EASA) is similarly responsible for organizations it certifies or oversees.¹⁸²

The occurrences are reported by individuals with key safety roles within aviation organizations, especially front-line professionals, such as pilots-in-command of an aircraft, persons connected to the safety management of airports and ground-handling service providers, air traffic controllers, etc.¹⁸³

Voluntary reporting systems

Voluntary reporting systems register information that is not included in the mandatory systems,¹⁸⁴ particularly latent conditions and human factor issues, such as inadequate procedures or regulatory gaps.¹⁸⁵ Experience has shown that accidents are often preceded by safety-related incidents and deficiencies, revealing the existence of safety hazards.¹⁸⁶

Voluntary reporting systems facilitate registration of information by persons who are not subject to mandatory reporting.¹⁸⁷

The duty to establish voluntary reporting systems applies to MS, aviation organizations, and EASA.¹⁸⁸ Data gathered must be forwarded

to the competent authority, namely the national regulator or EASA, depending on the certifying body.¹⁸⁹

In this regard, the EU has not followed the US approach. To enhance trust and encourage reporting, the US preferred the national voluntary reporting system to be established and managed by an independent third party, which has no regulatory or enforcement authority.¹⁹⁰

Content of the reports

To enable consistency and information sharing of information, occurrence reports must follow a uniform format, incorporating a minimum set of standardized data fields.¹⁹¹ Each occurrence reported should be risk-classified according to the common European risk-classification scheme for aviation.¹⁹² This enables consistent risk assessment and facilitates the identification of emerging safety trends at both national and EU levels.¹⁹³ The MS (or EASA, as the case may be) review and, if necessary, amend the classification of risk registered in the reports.¹⁹⁴ For less complex aircraft, the scheme adopts a simplified approach to maintain proportionality in regulatory burden.¹⁹⁵

Protection of information

Aviation safety is a responsibility shared among States, organizations, and individuals.¹⁹⁶ To ensure the continued availability of the reported data and information and encourage reporting, the data and information registered in occurrence reporting systems are protected through technical, organizational, and institutional means.¹⁹⁷ This enables proper and timely preventive actions to improve aviation safety.¹⁹⁸

Collection and storage

Each organization must appoint one or more qualified individuals to manage the collection, analysis, and storage of safety data independently.¹⁹⁹ Similarly, each MS has to designate one or more competent authorities to this end.²⁰⁰ Safety data is to be recorded in internal databases of each organization, while MS has to set up separate, central national databases, which collect the data recorded in

the databases under their jurisdiction, including the data from investigations of occurrences.²⁰¹ Small organizations may have simplified reporting mechanisms in place, by agreement with the competent authority.²⁰²

There are strict deadlines for submission of the reports, to expediate reporting and facilitate the recalling of facts and crucial details. Occurrences must be registered within 72 hours in the internal database of the respective aviation, save for exceptional circumstances.²⁰³ Following such registration, the organization has to report to EASA or the competent authority of each MS, as the case may be, within 72 hours of becoming aware of the occurrence.²⁰⁴

Aviation professionals, especially in the Air Traffic Management, have recognized the Regulation's contribution to safety improvement in an effort of the EU legislature to uncover as many occurrences as possible and establish EU-wide standardized reporting procedures and taxonomy of occurrences.²⁰⁵ Nevertheless, they have noted that the reporting requirements and procedures are not always sufficiently clear, while the tight deadlines may put aviation organizations under disproportionate time pressure and undermine the quality of the reports.²⁰⁶

Moreover, at the EU level, there is the European Central Repository (ECR), which stores all occurrence reports collected in the EU and is managed by the European Commission.²⁰⁷ The ECR registers occurrence reports from EASA and the MS, which have to transfer national occurrence reports to the ECR within 30 days from the date of entry in their national databases.²⁰⁸ Moreover, within 2 months from their storage in the national database, MS have to store in the ECR the information related to the analysis of the incident and the follow-up actions.²⁰⁹

‘Just culture’

Central to aviation safety is the concept of ‘just culture’,²¹⁰ where individuals are not punished for actions, omissions, or decisions taken by them that are commensurate with their experience and training, but in which gross negligence, wilful violations, and destructive acts are not tolerated.²¹¹ ‘Just culture’ recognizes that people usually set out to do

their best and achieve a good outcome: what seemed like the right thing to do in a situation to achieve efficiency and speed without compromising safety, may seem inappropriate in hindsight.²¹² Hence, 'just culture' contributes to building trust, cooperation, and understanding of the respective activities and responsibilities.²¹³

The core of just culture is its non-punitive approach. Data and information from reporting systems, particularly voluntary ones, should not be used to assign blame or liability.²¹⁴ Their sole purpose is improving aviation safety, while maintaining the confidentiality of the reporter and of the persons mentioned in occurrence reports.²¹⁵ The underlying rationale is that people are reluctant to provide safety information about their own errors or those of their colleagues, if they fear legal and disciplinary consequences.²¹⁶

Nevertheless, exceptions are allowed for behaviour that indicates wilful misconduct or gross negligence²¹⁷ and if the disclosure of data and information is necessary, ie not merely useful,²¹⁸ for: (i) the proper administration of justice or (ii) improving safety in general (eg for training and educational purposes).²¹⁹ In both cases, authorities must carefully consider any negative impact such disclosures might have on the future collection of safety data.²²⁰ Therefore, ICAO urges states to use authoritative safeguards like protective orders or closed proceedings to mitigate repercussions from disclosures.²²¹ In recognition of this fact, Regulation 376/2014 requires Member States to establish coordination mechanisms with judicial authorities, ensuring a proper balance between justice and the continuous availability of crucial safety information.²²²

At the level of aviation organizations, each organization is required to establish clear and concrete rules and procedures on the implementation of 'just culture',²²³ and inform all its personnel thereof in advance.²²⁴ To further guarantee compliance, each organization has to designate a monitoring body.²²⁵ At the same time, each Member State is mandated to designate a body responsible for the implementation of 'just culture', to which the staff of the organizations may directly report any suspected 'just culture' infringements.²²⁶

Confidentiality

Closely connected to 'just culture' is the protection of confidentiality. The reporters must be confident that their identity and the identity of their colleagues are protected, so that no disciplinary or legal measures can be instituted in response to their report.²²⁷ Confidential reporting implies that the personnel of the organization may bring information relevant to safety to the attention of a body, which is independent of management, and remain assured that their identity will be confidential to that body alone.²²⁸

EU law provides that, regarding internal reporting systems of aviation organizations, the reports disseminated should be disidentified and only the natural persons designated for the management of the data reported have full access to personal details.²²⁹ Databases held by MS and the EASA are not allowed to contain any personal details.²³⁰ Essential is the deidentification of the reports, not anonymity,²³¹ because it enables the analysts to contact the reporter for clarifications and enhances the credibility of reports.²³²

These rules do not prejudice the applicability of EU legislation on access to public documents²³³ and on personal data protection, especially Regulation 2018/1725²³⁴ and the GDPR.²³⁵

Occurrence analysis and follow-up

Safety information without analysis and follow-up actions is barely useful. Consequently, each aviation organization has to analyse the occurrences reported in its systems, to identify safety hazards. Thereafter, it has to determine any appropriate corrective or preventive action required to improve aviation safety,²³⁶ implement it in a timely manner, and monitor its implementation.²³⁷ The results of the safety analysis and the follow-up action have to be reported to the competent authority of that MS, or to EASA as the case may be, within 3 months from the notification of the occurrence, while a preliminary report has to be submitted within 30 days.²³⁸ MS and EASA are also under duty to analyse the occurrences, determine follow-up actions, and monitor their implementation.²³⁹

Exchange and dissemination of information

Exchange and dissemination of information, upon appropriate safeguards for the reporters, promotes safety across the entire aviation industry. It is often the case that the report of a single incident will prompt further reports and so provide a clearer indication of the scale of the problem.²⁴⁰ Hence, exchanging and disseminating safety information not only enable organizations to take advantage of the knowledge and experience of others but they also enable the identification of wider patterns and trends, and facilitate adoption of appropriate safety-improvement actions at the operational, organizational and regulatory level.²⁴¹

At the EU level, the European Central Repository (ECR) enables the exchange and dissemination of information among the MS, EASA, and the Commission.²⁴² This is operationalized through a network of safety analysts and a dedicated online platform, the European Co-ordination Centre for Mandatory Incident Reporting Systems (ECCAIRS),²⁴³ which uses the ICAO taxonomy of accident and incident reporting (ADREP).²⁴⁴

Full access to the ECR is granted to any entity with aviation safety regulatory competence.²⁴⁵ Indirect access, namely reports as a response to specific requests and upon evaluation of the request, may be provided to other aviation organizations and individuals (eg pilots, engineers, air traffic controllers, and safety researchers),²⁴⁶ under certain conditions²⁴⁷ and using a specific procedure, which involves national points of contact.²⁴⁸ Nonetheless, there is also the possibility of receiving aggregated or anonymised information, if the national point of contact finds that the requested information is unrelated to the interested party's own equipment, operations, or field of activity.²⁴⁹ In any case, all access requests and actions taken are recorded and transmitted to the Commission.²⁵⁰

To reinforce internal safety culture, each organization must regularly brief its personnel on safety trends and lessons learned.²⁵¹ Apart from knowledge dissemination, these briefings also demonstrate that reporting leads to meaningful action, thus encouraging personnel to report voluntarily incidents.²⁵²

Furthermore, each MS has to inform the public of the level of safety in civil aviation by publishing an annual safety review, which (i) summarises anonymised information on the type of occurrences and safety-related information reported through its national reporting systems, (ii) identifies trends, and (iii) identifies the action taken by the State.²⁵³ EASA is under a respective duty too.²⁵⁴ Such reports aim to inform the general public on the general level of aviation safety and must be drafted in plain and intelligible language.²⁵⁵

Investigation of occurrences

Safety reporting systems are related to the investigation of safety occurrences, which is governed by a detailed and multilayered framework under international and EU aviation law. This framework focuses on safety improvement through independent, non-punitive inquiry.

The value of regulating accident investigations to improve aviation safety was recognized already in the 1920s in some pieces of national legislation, such as in the UK and the US, partly as a response to high accident rates.²⁵⁶ During the drafting of the 1944 Chicago Convention, a special provision on accident investigation was included,²⁵⁷ while Article 37(k) thereof explicitly provided for the adoption of Standards and Recommended Practices (SARPS) on accident investigation. As a result, in 1951, the ICAO Council adopted Annex 13 'Aircraft Accident and Incident Investigation'.²⁵⁸ In the EU, Annex 13 is implemented through Regulation 996/2010, as amended.²⁵⁹

Annex 13 defines an 'investigation' as 'a process conducted for the purpose of accident prevention which includes the gathering and analysis of information, the drawing of conclusions, including the determination of causes and, when appropriate, the making of safety recommendations'.²⁶⁰ Investigations support the wider State Safety Programme, an integrated set of regulations and activities aimed at improving aviation safety, which each State has to establish and implement for the management of aviation safety.²⁶¹ The quality of the investigation of reported accidents, incidents, and safety issues largely determines the effectiveness of safety management systems.²⁶²

There are rules on international cooperation in the investigation of accidents, on the procedures to be followed in investigations, and on the preparation and publication of investigation reports.

Official investigation and international cooperation

States have to investigate accidents and serious incidents that have occurred in their territory and involve aircraft above a certain mass.²⁶³

However, they may delegate all or part of their duties to another State or to a specialized regional aviation safety organization by mutual agreement.²⁶⁴

It has been noted²⁶⁵ that the official investigation of serious incidents, apart from accidents, presents key opportunities to prevent future accidents. Investigating these events reveals how safety was compromised, why the situation did not escalate into an accident, and what was the influence of any barriers or mitigations that were effective. Besides, whereas accidents can bring political, legal, and emotional pressures, serious incidents often allow investigators to work in a calmer, more focused environment, with better access to evidence, since aircraft are typically intact, data are more easily recovered, and crew are available to share their accounts.

Apart from the State of Occurrence, other affected States may participate in the investigation, such as the State that has registered the aircraft (State of Registry), the national State of the Operator, the State where the particular aircraft type was designed (State of Design), and the State where the final assembly of the aircraft occurred (State of Manufacture).²⁶⁶ States of nationality of victims may also be involved.²⁶⁷ In the EU, the same goes for EASA as the overarching EU aviation safety authority.²⁶⁸ Special rules on the rights and duties of the participating States balance their legitimate interests with the need to preserve the integrity and coherence of the investigation.²⁶⁹

To strengthen cooperation across the EU, Regulation 996/2010 has established the European Network of Civil Aviation Safety Investigation Authorities (ENCASIA). Its goal is to support high-quality investigations and reinforce the independence of national authorities. ENCASIA also

promotes consistent standards in investigation methods and training for investigators.²⁷⁰

Guiding principles

The regulatory framework of occurrence investigations is shaped by two core principles. First, the sole objective of the investigation is to prevent future accidents or incidents, not to apportion blame or liability.²⁷¹ Second, States have to establish accident investigation authorities that are independent from both State aviation authorities and other entities that could interfere with the conduct or objectivity of the investigation, such as judicial authorities.²⁷² This independence must be both formal and substantive, with adequate financial, technical, and human resources ensured by the State.²⁷³ Independence in accident investigation not only facilitates the collection of evidence, since people hesitate to provide information about their own shortcomings when they fear legal or disciplinary repercussions,²⁷⁴ but it also underpins public and industry trust, mitigating the risk of institutional bias or cover-up.²⁷⁵

Scope and conduct of the investigation

In general, the investigation includes the collection, recording, and analysis of all available information; the issuance of safety recommendations, if appropriate; the determination of the causes, if possible; and the completion of a final report.²⁷⁶ The scope of each safety investigation is determined by the investigating authority, depending on the expected lessons for the improvement of aviation safety and accounting for the cost-efficient utilization of investigation resources in the EU.²⁷⁷ Although the investigation primarily focuses on the factors that are most likely to have influenced action, the dividing line between relevance and irrelevance is often blurred: seemingly unrelated data could later prove to be relevant once the relationship between the different elements of an occurrence is better understood.²⁷⁸

An aviation safety investigation begins with a report submitted under established occurrence reporting systems and unfolds through several phases.²⁷⁹ First, an investigation team is assembled, tailored in size and

expertise to the nature and severity of the event, with simpler cases often handled by a single investigator. The team is headed by the investigator-in-charge, a person appointed by the investigating authority, who has recognized expertise and experience, and is granted broad powers to lead the investigation.²⁸⁰

The investigative authority collects, secures, and preserves all relevant evidence, including physical materials and recorded data.²⁸¹ In this regard, coordination with judicial authorities may be necessary, although the two procedures are separate.²⁸² In any case, the investigating authority has unrestricted access to all evidence.²⁸³ Any statements, recordings, image and audio received, as well as the names of the persons involved in the occurrence, must remain confidential and not be released to the public.²⁸⁴

The evidence collected during an investigation helps to build a clear picture of the circumstances surrounding the occurrence and allows investigators to reconstruct the sequence of events, thus leading to the identification of both causal and contributory factors.²⁸⁵ Although investigation authorities around the world may use different methodologies, or often a mix of them, depending on the specific case, all approaches consider systemic factors as part of the analysis.²⁸⁶ The final phase involves analysing the information to uncover technical, operational, and organizational issues, leading to conclusions and recommended safety actions to prevent recurrence. These are laid down in the investigation reports.

Investigation reports

Special rules govern the preparation and publication of investigation reports, including the safety recommendations of such reports.

Duty to prepare and publish investigation reports

Within 30 days from the occurrence, the investigating State has to transmit a preliminary report to the other participating States and to ICAO.²⁸⁷

Draft reports remain confidential²⁸⁸ and, prior to their finalization, the investigating State has to invite comments from the other participating

States.²⁸⁹ The final report should be publicly released online not later than 12 months after the occurrence, if possible.²⁹⁰

The reports have a specified structure according to ICAO templates. They must include detailed information regarding the facts of the accident, their analysis, and the conclusions.²⁹¹ The conclusions comprise a list of the findings, as well as the causes and/or contributing factors established in the investigation, both immediate and systemic ones.²⁹² 'Causes' are defined as 'actions, omissions, events, conditions, or a combination thereof, which led to the accident or incident', while 'contributing factors' are 'actions, omissions, events, conditions, or a combination thereof, which, if eliminated, avoided or absent, would have reduced the probability of the accident or incident occurring, or mitigated the severity of the consequences of the accident or incident'.²⁹³

The investigation reports need to reflect the independence of the investigating body: they have to demonstrate that all circumstances were objectively examined, including systemic causes, such as company management, the regulatory framework, and sometimes the Government Department responsible for transport policy.²⁹⁴

Safety recommendations

Where appropriate, investigation reports should formulate safety recommendations, which are central to the preventive function of the inquiry.²⁹⁵ These recommendations are non-binding and do not create any presumption of fault or legal responsibility.²⁹⁶

The investigating authority is required to issue safety recommendations to the appropriate authorities, whether domestic or foreign, capable of implementing remedial action.²⁹⁷ Participating States may issue recommendations as well. ICAO encourages inter-state coordination before the release of such recommendations, to ensure appropriateness and effectiveness of the recommendations.²⁹⁸

In addition, States may issue Safety Recommendations of Global Concern (SRGCs), which regard systemic deficiencies having a probability of recurrence, with significant consequences at a global level, and requiring timely actions to improve safety.²⁹⁹ SRGCs are

submitted to ICAO and entered into a publicly accessible global database, ensuring international transparency and collective responsiveness.³⁰⁰

Recipient States are obliged to respond within 90 days, indicating either the measures taken (or planned) to implement the recommendation, or the reasons for inaction.³⁰¹ The investigating authority must document these responses and assess their adequacy, providing justification if it disagrees with a decision to take no action.³⁰² Furthermore, all authorities or entities receiving recommendations are expected to monitor and record in the European Central Repository the progress of their implementation.³⁰³

Overall evaluation

The aviation occurrence reporting and investigation system has proven highly successful. Its key elements are a constant collection and analysis of safety data and information at all levels and its dissemination to all stakeholders. At the same time, safety is both a shared core responsibility and concern across the industry and policymakers. The concept of 'just culture' is a major contributor to safety, despite the challenges in its practical implementation. The international regulation of occurrence reporting and investigation, a practical necessity emanating from the nature of aviation operations, has led to the creation of global standards and has hugely facilitated risk management and mitigation. Despite the inherent risk of air transport, many estimates consider aviation as the safest mode of transportation. This is underlined by a great amount of public trust: it is not accidental that most air travelers are concerned with delays rather than death or physical injuries. There are always challenges,³⁰⁴ as recent aviation accidents have shown, but sharing information and experiences across the industry and at an international level significantly facilitates the devise of appropriate solutions.³⁰⁵

DRAWING FROM THE AVIATION EXPERIENCE TO IMPROVE THE AIA REPORTING SYSTEM

As already shown, aviation can provide useful insights for the improvement of the incident reporting and investigation system of the AIA, given the parallels between the two sectors. However, despite their similarities, caution is required.

Differences between AI systems and aviation

Before exploring possible measures for the AI industry, it is important to consider the key differences that set it apart from aviation.

Aviation is a highly developed industry. It is heavily regulated based on lessons learned during many decades of experience, including dozens of fatal accidents. Safety regulation is largely standardized at the global level, which reduces competitive pressures on safety issues.

Furthermore, it is founded on well-established technical principles and a highly developed safety culture. From a technological perspective, aviation is a relatively slow-evolving industry, which gives actors the time to adjust to technological developments and related challenges. Moreover, aviation occurrences take place mainly in the physical world. Accidents in aviation are immediately apparent and may significantly affect the risk perception of passengers,³⁰⁶ which means that the reputational impact of accidents can be huge.³⁰⁷ Closely connected is the financial impact of accidents, which can take various forms, such as civil liability,³⁰⁸ administrative fines and criminal charges,³⁰⁹ loss of revenue, costs for repairs, and modifications etc. The multifaceted repercussions of accidents result in a convergence of interests of the various stakeholders, driving them to prioritize safety.³¹⁰

The AI industry is still at its beginning. AI systems are used across a wide spectrum of industries, whose stakeholders may have diverse interests. The regulation of the AI industry greatly varies among States, while the experience with accidents is very limited. AI-related accidents may not become widely known, especially if they are constrained in the digital world, while their reputational effect can be less severe. At the same time, however, AI-related accidents or malfunctions may have a

wider societal impact, despite any lack of fatalities.³¹¹ Moreover, the safety culture of AI providers is still in early developmental stages. This is related to the strong competition among AI system developers combined with the rapid evolution of technology. In this context, safety is balanced against technological advancement and market dominance.³¹² Strategic competition among States further complicates things.³¹³ Thus, safety concerns in product development and deployment may not receive appropriate attention.³¹⁴ A related factor is that AI systems are opaque and non-deterministic, which means that it is challenging to establish clear causal connections between human behaviour in the development or operation of the AI system and the output or behaviour of the AI system.³¹⁵ Furthermore, the development and operation of AI systems are often protected by intellectual property rights, which need to be taken into account when incidents are reported and investigated.³¹⁶

Another major element is that aviation safety regulation has been strongly internationalized since 1944 (when the Chicago Convention was signed), adopting a top-down approach, where rules are decided mainly at the ICAO level and then implemented by States. On the contrary, AI safety regulation has so far been a national concern, or a regional concern in the case of the EU, with different States having different priorities and adopting often divergent regulatory approaches, ranging from comprehensive, proactive regulation in the EU, such as the AIA, to a hands-off and deregulatory approach at the US federal level.³¹⁷

Nevertheless, aviation rules and practices can inspire enhancements in the reporting and investigation of AI system incidents.

Establish a strong safety culture with appropriate safeguards

Safety reporting systems are intricately connected to a wider safety culture, which includes the willingness to communicate safety issues.³¹⁸ The development of such a culture has significantly improved aviation safety.

Alongside its business management dimensions,³¹⁹ a safety culture entails that organizations and natural persons feel safe to report safety-related information.³²⁰ Especially when reporters are natural persons, personal data should be removed from incident reports. Information contained in the reports should be strictly confidential and used exclusively to promote safety. This means that safety reporting should be disconnected from liability. Important is the existence of a 'just culture', in which natural persons and organizations are not punished for behaviour that can be deemed reasonable under the circumstances, but in which recklessness is not acceptable.³²¹ It is encouraging that the Code of Practice for GP AI models already points to this direction.³²²

Establish voluntary reporting systems

Voluntary reporting systems often capture safety data and information not covered by the mandatory reporting systems and can be extremely useful, as the experience in aviation has shown.

All organizations in the supply chain of AI systems should establish voluntary reporting systems. For providers of general-purpose AI models with systemic risk, such a duty seems to already exist under the Code of Practice for GP AI models as a post-monitoring duty.³²³ In a similar vein, the Commission could issue guidelines under Article 95 AIA, in which it will be clarified that the post-monitoring duties of AI system providers comprise the establishment of voluntary reporting systems, which would also permit the submission of anonymous reports.

For voluntary reporting systems to be effective, a strong safety culture is needed, as already mentioned. This requires confidentiality of information and strong safeguards for the reporters, including de-identification of the reports and the possibility of anonymous reporting. Direct access to the reported data and information should be limited only to specific persons within each organization or entity, to supervisory authorities, and the AI Office.

In voluntary reporting systems, there should be the possibility to register potential accidents or 'hazards' under the OECD terminology. In other words, it should suffice that the event could have resulted in harm

or damage (near miss), since it is often a matter of luck whether actual harm or damage occurred.³²⁴

In addition, it should be possible for every professional to report an incident in an official central database, even if the incident has not been reported in the internal database of the organization concerned. The reasons are related, on the one hand, to the possibility that organizations may be tempted to suppress useful safety information, especially if it puts them in an unfavourable light or at a competitive disadvantage; on the other hand, to psychological encouragement of the reporters, who for various reasons may be reluctant to report incidents internally.³²⁵ Besides, the possibility of anonymous reporting in a central database has been identified as desirable for reporting organizations too, since it protects them from reputational repercussions.³²⁶

It would enhance trust and facilitate reporting, if such central database was held by a neutral, independent third party, namely an organization with no regulatory or enforcement tasks.³²⁷ This role could be fulfilled by a public or private organization with technical expertise, such as an academic or technology institution. Although the EU has not followed this approach in aviation, it would be worth implementing it in the context of AI systems, where the challenges of regulatory compliance and the related fear of sanctions appear to be bigger compared to aviation.

It is noteworthy that some mechanisms for the establishment of voluntary reporting systems are already in place, in transposition of the requirements of the Whistleblower Directive.³²⁸ Although the scope of such systems under the Whistleblower Directive is narrower, their existence alleviates the administrative burden of establishing voluntary reporting systems with a wider scope.

Clarify and extend incident reporting duties

The proper function of reporting systems requires clear rules on *who* must report *what*, *to whom* and *when*,³²⁹ alongside simple procedures.

Clarify ‘what’

As in aviation, a detailed taxonomy of ‘risk’ and ‘harm’ would contribute to a shared understanding of these concepts across the EU. A common taxonomy of risks significantly facilitates organizations and supervisory authorities in risk assessment and analysis, including the identification of common issues and trends³³⁰ while helping them determine appropriate preventive and corrective measures.

Similar to aviation legislation, which distinguishes between actual occurrences (‘accidents’) and potential occurrences (‘incidents’), it would be useful to have reporting systems cover potential harm, injury, or disruption too. Currently, the incident reporting duties under the AIA regard only actual harm, injury, or disruption.³³¹ Nonetheless, voluntary reporting systems can be established to include information about ‘near-misses’. To denote potential harm and specify reporting duties, the OECD draft definitions of ‘AI hazard’ and ‘serious AI hazard’ could be used as a reference. These are distinct from an ‘AI incident’, a ‘serious AI incident’, and an ‘AI disaster’, which denote actual harm.³³²

Moreover, a common taxonomy of harm would help to determine the required degree of ‘seriousness’ of harm to a person’s health, property, or the environment, as well as of the ‘seriousness’ and ‘irreversibility’ of disrupting the management or operation of critical infrastructure, which are mentioned in the definition of ‘incident’ under the AIA. Although determining ‘seriousness’ for so many diverse cases is a highly complicated and challenging task,³³³ it is necessary for the proper implementation of the AIA.

In addition, the enigmatic ‘infringement of obligations under Union law intended to protect fundamental rights’ needs to be further clarified through official guidance. A comprehensive, albeit not exhaustive, list of such obligations should be created and a clarification of who should establish the (potential) ‘infringement’ and how.

The above can be complemented by standardized reporting templates. Standardization of the reported information contributes to uniform implementation and easier processing, including meaningful data aggregation and comparison³³⁴ and has proven particularly useful in aviation.

Incident reports should contain enough information to enable the determination of the nature, gravity, and potential causes of the incident.³³⁵ The AI Office has been mandated to develop such templates.³³⁶ It could take into account elements of reporting frameworks already developed by other organizations, such as the OECD and the Center for Security and Technology (CSET).³³⁷

Clarify and extend ‘who’

Following the example of aviation, the personnel (employees and contractors) of providers and deployers of all AI systems should be under duty to register serious incidents to internal safety reporting systems.

The current provisions of the AIA on providers and deployers of high-risk AI systems entail such duty. Concerning providers of high-risk AI systems, internal reporting obligations are necessary in practice for them to comply with their reporting duties under Article 73 AIA. Moreover, they can be seen as part of the existing duties of providers of high-risk AI systems to establish, document, implement, and maintain a risk-management system,³³⁸ as well as to establish and document a post-marketing monitoring system.³³⁹ Similar obligations are implied for the personnel of deployers of high-risk AI systems regarding their notification duties under Articles 26(5) and 73 AIA.

Furthermore, since safety is among the core objectives of the AIA,³⁴⁰ incident reporting duties could extend to all developers and deployers of AI systems, whether they are classified as high-risk or not. Otherwise, potentially valuable information and experiences might go unnoticed. To avoid disproportionate burdens on small and medium enterprises (SMEs), simplified reporting procedures could be established, eg through the requirement of less information and longer deadlines to report. In aviation, such lighter reporting duties already exist.³⁴¹

Ensure reporting systems are accessible and easy to use

Reporting systems, whether mandatory or voluntary, need to be user-friendly and easily accessible.³⁴² The aviation experience has shown that the reporting procedures should be simple, straightforward, and help the user to report all valuable information regarding the incident, like what happened, when, under what circumstances, and what the harmful outcome was or could have been.³⁴³

Set up mandatory incident investigation by an independent authority

Mandatory accident investigation by an independent authority has proven very useful in aviation. It would be recommended to establish respective provisions in the AIA too. Currently, the AIA provides for incident investigation by the developer or deployer of high-risk AI systems, while national supervisory authorities may subsequently investigate and verify the compliance of the respective organization with the AIA.³⁴⁴ This system is prone to create conflicts of interests in the incident investigation. In the context of AI, an independent investigation body can identify failed safeguards, assess existing policies, and gather data on affected populations and risk factors—information which may reveal the likelihood of past, unreported, or future incidents.³⁴⁵

It should be noted that mandatory incident investigation by an independent authority already applies to AI systems used in the transport sector, under the sectoral EU legislation.³⁴⁶ However, establishing mandatory investigation of AI-related safety occurrences in other sectors is complicated.

Challenges regarding AI systems

There are significant challenges in establishing an aviation-like system of mandatory incident investigation by an independent authority.

AI systems are much more diverse than aviation products and systems. They are widely used in many sectors, and their use keeps expanding,

which means that the number of incidents can be significantly larger than the number of aviation safety occurrences. This could not only strain the resources available for independent incident investigation but also be regarded as too onerous for developers and deployers of AI systems.

Additionally, the technology of AI systems is new and constantly evolving at a rapid pace, which renders it difficult to establish standardized methods and procedures. Besides, the opacity of commonly used machine-learning systems poses a significant barrier to external investigation, particularly when investigators lack detailed knowledge of the specific AI technology being employed.

Moreover, AI systems are often protected by intellectual property rights (IPR), especially trade secrets, which could be compromised during an independent investigation. An additional, complicating factor is the difficulty of recruiting AI experts to act as independent investigators. First, such experts are hard to attract to an independent authority, given the intense competition among AI companies for talent.³⁴⁷ Second, these are likely to have ties with other high-tech companies or envision a career shift to industry ('revolving doors'),³⁴⁸ which might compromise the independence of the investigation.

Overcoming challenges

These challenges are not insurmountable.

The scope of mandatory independent incident investigation could be limited to certain categories of incidents. Mandatory independent investigation might only apply to cases of 'serious incidents' under Article 3(49)(a), (b), and (d), ie cases of death or serious harm to a person's health, serious and irreversible disruption of the management or operation of critical infrastructure, and serious harm to property or the environment. These are cases expected to occur less often and will probably be investigated either way by safety and judicial authorities. In addition, they are more straightforward to implement than cases of Article 3(49)(c), ie infringement of obligations under Union law intended to protect fundamental rights.

It should be underlined that incident investigation is already mandatory under both the AIA and many pieces of sectoral legislation,³⁴⁹ despite the novelty and the rapid evolution of technology. Investigations will be conducted either way. Thus, first, it is a matter of time before standardized methods and procedures are developed in the form of EU harmonized standards, codes of practice, or Commission guidelines.³⁵⁰ Second, to enhance safety, we need to avoid a system prone to creating conflicts of interests, namely a system in which incidents are investigated by the respective organizations themselves.

Explainability of AI systems is already a basic requirement under the AIA.³⁵¹ It is also one of the core principles of ethical AI³⁵² and a central element of social acceptance of such technology.³⁵³ As a result, significant related research is undertaken, and explainability issues will be gradually overcome.

Protection of IPR in the context of supervision and enforcement of the AIA by public authorities is already accounted for in the AIA.³⁵⁴ Besides, IPR protection of AI systems is not unconditional; it is balanced against the public interest in safety.³⁵⁵

Finding independent technical experts is an issue that the AIA has dealt with through the establishment of the Scientific Panel of Independent Experts,³⁵⁶ to which every MS may have access.³⁵⁷ Additional measures at the national level may be envisaged, like economic, reputational and personal-growth incentives.³⁵⁸ In aviation, one of the methods used to resolve such issues are agreements on the delegation of the investigation duties to the competent authority of another State, which may have more resources and greater expertise.³⁵⁹

The phenomenon of ‘revolving doors’, ie employees of public organizations, especially regulators, shifting to posts in the industry and vice versa, is an unavoidable risk. It can be mitigated through appropriate legislative restrictions, such as cooling-off periods after termination of employment in the public sector.³⁶⁰ At the same time, previous employment in the industry can be an asset for accident investigators because it provides them with experience on how things work in practice, while an industry employee or contractor with experience in independent accident investigation can help the

organization improve its safety approach and safety culture. Besides, the risk of revolving doors is not very likely to influence the independence of the investigation, if accident investigation authorities are independent from regulators and have no authority to impose sanctions. This also explains the lack of reported pertinent issues in aviation.

Thus, the challenges in setting up a mandatory incident investigation by an independent authority can be overcome or significantly mitigated.

Provide safety recommendations and ensure follow-up

One of the most useful elements in aviation accident reports is the safety recommendations addressed to various actors and the procedure for follow-up actions.

The Commission could clarify that the AI incident investigation reports should include such recommendations, where possible. This has already been recognized by the Code of Practice for providers of GP models with systemic risk.³⁶¹ At the same time, a follow-up procedure should be formally established, for the addressees of the recommendations to act upon the lessons learned. This would not only promote safety but also enhance accountability, which is a significant parameter of ethical AI.³⁶² Besides, recording real-world responses to incidents helps AI engineers understand and identify ethical paths forward after an incident.³⁶³

Organize effective dissemination of safety information

Effective dissemination of safety information is crucial for the safety improvement of AI systems. Among others, it enables identification of risk patterns and facilitates cross-industry collaboration.³⁶⁴

It would be very useful to set up and maintain a European Central Repository for AI incidents, facilitate information sharing among the national supervisory authorities and the AI Office, and provide tailored safety information to parties with a legitimate interest. The Scientific

Panel of Independent Experts³⁶⁵ could assist the AI Office in the analysis of the safety information. Moreover, AI systems could be used to help process the information gathered, provide summaries of varying detail for different purposes, visualize safety data, identify systemic trends, etc.

It should not be overseen that the dissemination of information, when undertaken with the appropriate safeguards for reporters, enhances trust in the safety system by professionals, thus encouraging voluntary incident reporting.

In addition, it would be useful to lay down a duty for MS and the AI Office to issue yearly public reports on the status of AI safety. The reports would include aggregated and anonymised information on AI incidents, follow-up actions, statistics, and trends. This will enhance overall transparency and facilitate public trust and acceptance of AI systems.

CONCLUSION

In conclusion, safety reporting systems are crucial for taking advantage of the great opportunities offered by AI systems while mitigating related risks. They allow lessons to be drawn from experience, help the AI industry develop, and assist society to mitigate risks. From an ethical perspective, incident reporting contributes to the development of more trustworthy AI systems, especially by enhancing technical robustness and transparency.³⁶⁶

Even though the current provisions of the AIA on incident reporting and investigation are a step in the right direction, they have considerable room for improvement. The experience in aviation can be utilized to improve safety reporting, yet account should be taken for the differences between the aviation and the AI industries.

A series of measures could significantly improve the current system: (i) clarification of the reporting duties, including the development of a common taxonomy of risk or harm, alongside the expansion of the persons and entities under duty; (ii) disconnection of incident reporting

and investigation from liability; (iii) establishment of voluntary reporting systems; (iv) provisions on confidentiality of the reports and protection of the reporters; (v) mandatory investigation of at least some serious incidents by an independent authority; (vi) effective dissemination of safety information. The existing framework of the AIA, through implementing instruments or through official guidelines, could accommodate some of these elements, such as the clarification of the reporting duties, the establishment of user-friendly voluntary reporting systems, and the effective dissemination of safety information. Other proposals would require legislative intervention, such as the mandatory investigation of incidents by an independent authority.

However, the most important factor is the creation of a strong safety culture in both organizations and regulators. Such a culture entails that safety is seen by all persons and entities involved, including States and supervisory authorities, as a shared responsibility of utmost importance.³⁶⁷

Overall, as AI systems become increasingly integrated into almost all sectors of human activity, effective incident reporting contributes to a broader understanding of AI risks and is a significant step towards trustworthy AI. Aviation can provide valuable guidance to this end. Its example has demonstrated that industries can grow rapidly without compromising safety if they prioritize safety and commit to systematic and structured safety learning.

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Footnotes

- 1 Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence and amending Regulations (EC) No 300/2008, (EU) No 167/2013, (EU) No 168/2013, (EU) 2018/858, (EU) 2018/1139 and (EU) 2019/2144 and Directives 2014/90/EU, (EU) 2016/797 and (EU) 2020/1828 [2024] OJ L2024/1689.
- 2 AIA, art 1(1).
- 3 AIA, art 5.
- 4 AIA, arts 6–49.
- 5 AIA, art 50.
- 6 AIA, arts 51–55.
- 7 AIA, art 2(1).
- 8 Matthias Samwald and others, ‘Code of Practice for General-purpose AI Models. Safety and Security Chapter’
<<https://ec.europa.eu/newsroom/dae/redirection/document/118119>>
accessed 28 July 2025 (Code of Practice for GP AI models).
- 9 See AIA, arts 55(2), 56 and recital (117).
- 10 Sarah Hartmann, ‘Artikel 73 Meldung schwerwiegender Vorfälle’, in Mario Martini and Christiane Wendehorst (eds), *KI-VO Verordnung zur Künstliche Intelligenz* (C.H. Beck 2024) para 3; Christian Djefal, ‘KI-VO Artikel 73 Meldung schwerwiegender Vorfälle’, in Jens Schefzig and Robert Kilian (eds), *BeckOK KI-Recht* (2nd edn, C.H. Beck 2025, updated 1.May 2025) paras 27-28; Clemens Bernsteiner and Thomas Rainer Schmitt, ‘Art. 55 Pflichten der Anbieter von KI-Modellen mit allgemeinem Verwendungszweck mit systemischem Risiko’ in Mario Martini and Christiane Wendehorst (eds), *KI-VO Verordnung zur Künstliche Intelligenz* (C.H. Beck 2024) para 15.
- 11 Regulation (EU) 2017/745 of the European Parliament and of the Council of 5 April 2017 on medical devices, amending Directive 2001/83/EC, Regulation (EC) No 178/2002 and Regulation (EC) No 1223/2009 and repealing Council Directives 90/385/EEC and 93/42/EEC [2017] OJ L117/1, arts 87 ff.

- 12 Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC [2016] OJ L119/1, art 33.
- 13 Commission, 'Draft Guidance Article 73 AI Act – Incident reporting' 26 September 2025 <<https://digital-strategy.ec.europa.eu/en/consultations/ai-act-commission-issues-draft-guidance-and-reporting-template-serious-ai-incidents-and-seeks>> accessed 17 October 2025, para (3).
- 14 Directive (EU) 2019/1937 of the European Parliament and of the Council of 23 October 2019 on the protection of persons who report breaches of Union law [2019] OJ L305/17 (Whistleblower Directive).
- 15 AIA, recital (155).
- 16 AIA, arts 73(1) and 26(5).
- 17 AIA, art 17(1)(i).
- 18 AIA, art 73(1).
- 19 AIA, art 60(7).
- 20 AIA, art 55(1)(c).
- 21 AIA, recital (155).
- 22 *ibid.*
- 23 Commission, 'Draft Guidance Article 73 AI Act' (n13) para (2).
- 24 Djefal (n 10) para 1; Ayça Atabey, Lachlan Urquhart and Buckhard Schafer 'Article 73 Reporting of Serious Incidents' in Ceyhun Necati Pehlivan, Nikolaus Forgó and Peggy Valcke (eds), *The EU Artificial Intelligence (AI) Act: A Commentary* (Wolters Kluwer International 2024), para 1.
- 25 AIA, art 73(1).
- 26 AIA, art 3(3).
- 27 'Deployer' means a natural or legal person, public authority, agency, or other body using an AI system under its authority except where the AI

system is used in the course of a personal non-professional activity (AIA, art 3(4)).

28 AIA, art 26(5), third sentence.

29 AIA, art 26(5), fourth sentence.

30 AIA, art 73(5).

31 AIA, art 55(1)(c).

32 AIA, art 3(3).

33 AIA, art 3(66).

34 AIA, art 3(64).

35 AIA, art 3(65).

36 AIA, art 51(2).

37 AIA, art 51(1)(b).

38 AIA, art 3(49).

39 Code of Practice for GP AI models, Measure 9.3 Reporting timelines; Bernsteiner and Schmitt (n 10), para 17; Schneider, KI-VO Art. 55 Pflichten der Anbieter von KI-Modellen mit allgemeinem Verwendungszweck mit systemischem Risiko' in Jens Schefzig and Robert Kilian (eds), *BeckOK KI-Recht* (2nd edn, C.H. Beck 2025, updated 1 May 2025) para 12.

40 Code of Practice for GP AI models, Measure 9.3 Reporting timelines.

41 Commission 'Draft Guidance Article 73 AI Act' (n13) para (11).

42 Ibid, para (10).

43 OECD, 'Defining AI Incidents and Related Terms', OECD Artificial Intelligence Papers, No. 16 (OECD Publishing 2024) <<https://doi.org/10.1787/d1a8d965-en>> accessed 24 June 2025, 11. A representative of the European Commission co-chaired the OECD expert group on AI incidents, as acknowledged in page 3 of the OECD paper.

44 The Council deleted the words 'might have lead or might lead' from the

initial Commission proposal, which had been retained by the Parliament—
see the different versions of art 3(49) in the legislative process under
<https://assets.ctfassets.net/mj324dykhxwi/5WoflzmPRcNsBe0ommdIMG/383e84b1508e2b379b3afa754b15cdbc/AIAct_final_four-column21012024.pdf> accessed 24 June 2025, 312–313.

- 45 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para (8); Lukas Feiler and Beat König ‘Article 3 Definitions’ in Ceyhun Necati Pehlivan, Nikolaus Forgó and Peggy Valcke (eds), *The EU Artificial Intelligence (AI) Act: A Commentary* (Wolters Kluwer International 2024) para 3.49.
- 46 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para (15); Code of Practice for GP AI models, Measure 9.3 Reporting timelines. See also OECD (n 43) 11, which clarifies that psychological harms and harms to mental health are included under the broader concept of health. Compare the definition of ‘health’ by the World Health Organisation ‘Health is a state of complete physical, mental and social well-being and not merely the absence of disease or infirmity’ in the preamble of its Constitution, 14 UNTS 115, as amended.
- 47 See, for example, Noor Al-Sibai, ‘ChatGPT Is Telling People With Psychiatric Problems to Go Off Their Meds’ (*Futurism* 14 June 2025)
<<https://futurism.com/chatgpt-mental-illness-medications>> accessed 24 June 2025. Compare ‘Emotional risks of AI companions demand attention’ (2025) 7 *Nat Mach Intell*, 981–982; Zara Abrams, ‘Using Generic AI Chatbots for Mental Health Support: A Dangerous Trend’ (American Psychological Association, Services 12 March 2025)
<<https://www.apaservices.org/practice/business/technology/artificial-intelligence-chatbots-therapists>> accessed 24 June 2025.
- 48 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para (15), which mentions as examples a condition necessitating prolongation of existing hospitalization or a clinically relevant increase in the duration of a surgical procedure.
- 49 AIA, art 2(62) combined with art 2(4) of the Directive (EU) 2022/2557 of the European Parliament and of the Council of 14 December 2022 on the resilience of critical entities [2022] OJ L333/164.
- 50 Directive 2022/2557, art 2(5).

- 51 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para 19.
- 52 Id., para 21.
- 53 Christiane Wendehorst, ‘Artikel 3 Begriffsbestimmungen’ in Mario Martini and Christiane Wendehorst (eds), *KI-VO Verordnung zur Künstliche Intelligenz* (Beck 2024), para 335.
- 54 ibid, para 337; Commission ‘Draft Guidance Article 73 AI Act’ (n13) para (27), which notes that in any case the damage should exceed 5% of the purchased price.
- 55 Directive 2004/35/EC on environmental liability with regard to the prevention and remedying of environmental damage [2004] OJ L143/65.
- 56 Directive (EU) 2024/1203 on the protection of the environment through criminal law [2024] OJ L 1203.
- 57 Commission ‘Draft Guidance Article 73 AI Act’ (n13) paras 28–29.
- 58 Id., para 30.
- 59 This requires that the EU has competency to legislate on issues of outer space, which might be disputed in view of art 189(2) TFEU. However, the Commission, also in its guidance on art 73 AIA, seems to accept such competence, since it mentions the destruction of a satellite as an example of destruction of critical infrastructure—see Commission ‘Draft Guidance Article 73 AI Act’ (n13) para 21.
- 60 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para 23, which cites recital (1) AIA.
- 61 Id., paras 24–25; Wendehorst (n53), para 336.
- 62 Commission ‘Draft Guidance Article 73 AI Act’ (n13) para 26.
- 63 AIA, art 55(1)(d) and recital (114) last sentence.
- 64 See US National Institute for Standards and Technology (NIST), ‘Artificial Intelligence Risk Management Framework: Generative Artificial Intelligence Profile’ (July 2024) <<https://doi.org/10.6028/NIST.AI.600-1>> accessed 24 June 2025, para 2.9. In data poisoning, an adversary compromises a training dataset used by a model to manipulate its outputs or operation, while

prompt injection is the direct or indirect modification of the input is provided to the model so that it behaves in unintended ways.

- 65 Commission, 'Draft Guidance Article 73 AI Act' (n13) para (13).
- 66 Id.
- 67 Id. para (14).
- 68 See Nadin Yusif, 'Parents of teenager who took his own life sue OpenAI', BBC 27 August 2025 <<https://www.bbc.com/news/articles/cgerwp7rdlvo>> accessed 21 October 2025; Daniel Atherton, 'Incident Number 1026: Multiple LLMs Allegedly Endorsed Suicide as a Viable Option During Non-Adversarial Mental Health Venting Session.', 12 April 2025, in Daniel Atherton (ed.) *Artificial Intelligence Incident Database*, <incidentdatabase.ai/cite/1026> accessed 21 October 2025.
- 69 Djefall (n 10), para 3.
- 70 AIA, art 73(2).
- 71 ibid para 14.
- 72 AIA, art 73(2).
- 73 AIA, art 73(2), second sentence.
- 74 See Djefall (n 10) para 42.
- 75 AIA, art 73(4).
- 76 AIA, art 3(61) defines a 'widespread infringement' as any act or omission contrary to Union law protecting the interest of individuals, which (i) has harmed or is likely to harm the collective interests of individuals residing in at least two Member States other than the Member State in which (a) the act or omission originated or took place; (b) the provider concerned, or, where applicable, its authorized representative is located or established; or (c) the deployer is established, when the infringement is committed by the deployer; (ii) has caused, causes or is likely to cause harm to the collective interests of individuals and has common features, including the same unlawful practice or the same interest being infringed, and is occurring

concurrently, committed by the same operator, in at least three Member States.

77 As these examples are mentioned by the Commission 'Draft Guidance Article 73 AI Act' (n13) para 35.

78 AIA, art 73(3).

79 AIA, art 73(2).

80 AIA, art 73(4).

81 AIA, art 73(3).

82 See also Djefal (n 10), para 48, who suggests the application of paragraph (4) by analogy, ie a suspicion of a causal link, to avoid unfounded reports with respective costs and reputational damage for the AI provider.

83 AIA, art 73(5).

84 Djefal (n 10), para 53; Atabey and others (n 24), para 3.5.

85 Above n 8, Measure 9.3 Reporting Timelines.

86 *ibid.*

87 *ibid.*

88 *ibid.*

89 *ibid.*, Measure 9.4 Retention period.

90 See Djefal (n 10), para 36.

91 Compare Regulation (EU) 2019/1020 of the European Parliament and of the Council of 20 June 2019 on market surveillance and compliance of products [2020] OJ L169/1 (Market Surveillance Regulation), art 20(3), to which art 73(11) AIA explicitly refers.

92 AIA, art 55(1)(c).

93 AIA, art 73(1).

94 See also Commission 'Draft Guidance Article 73 AI Act' (n13) para 37, which mentions the business location of the deployer.

- 95 Compare Regulation (EC) No 864/2007 of the European Parliament and of the Council of 11 July 2007 on the law applicable to non-contractual obligations (Rome II) [2007] OJ L199/40, art 4(1).
- 96 See AIA, art 3(47).
- 97 AIA, art 55(1)(c).
- 98 AIA, art 70(1).
- 99 Djefal (n 10), para 67; Atabey and others (n 24), para 3.9.
- 100 These are AI systems used in (i) biometrics, (ii) critical infrastructure, (iii) education and vocational training, (iv) employment, workers' management and access to self-employment, (v) access to and enjoyment of essential private services and essential public services and benefits, (vi) law enforcement, (vii) migration, asylum and border control management, and (viii) administration of justice and democratic processes.
- 101 See, however, Djefal (n 10), para 69, who suggests that there is 'equivalence' under two cumulative requirements: (i) the reporting should concern causal relationships between the system and the incident, and (ii) there must be a specific minimum deadline for reporting.
- 102 Above n 11.
- 103 Regulation (EU) 2017/746 of the European Parliament and of the Council of 5 April 2017 on *in vitro* diagnostic medical devices and repealing Directive 98/79/EC and Commission Decision 2010/227/EU [2017] OJ L11/176.
- 104 AIA, art 73(10).
- 105 Commission 'Draft Guidance Article 73 AI Act' (n13) para 67.
- 106 AIA, art 73(6).
- 107 *ibid*.
- 108 Djefal (n 10), para 56.
- 109 *ibid*, para 57.
- 110 *ibid*, para 57.

- 111 *ibid.*
- 112 *ibid.* A ‘notified body’ means a conformity assessment body notified in accordance with this Regulation and other relevant Union harmonisation legislation (AIA, art 3(22)). A ‘conformity assessment body’ means a body that performs third-party conformity assessment activities, including testing, certification, and inspection (AIA, art 3(21)).
- 113 AIA, art 73(6).
- 114 Djeflal (n 10), paras 58–59.
- 115 Commission, ‘Draft Guidance Article 73 AI Act’ (n13) para 43.
- 116 AIA, art 60(7).
- 117 *ibid.*
- 118 AIA, art 73(8) combined with Market Surveillance Regulation (Regulation (EU) 2019/1020 of the European Parliament and of the Council of 20 June 2019 on market surveillance and compliance of products [2020] OJ L169/1) art 19(1).
- 119 *ibid.*
- 120 AIA, art 76(3).
- 121 See AIA, arts 99 and 101.
- 122 See Javier Espinoza, ‘Why big-tech fines do not work’ (*Financial Times* 10 April 2024) <<https://www.ft.com/content/ba6eb664-b981-42d7-b24a-65e7e19889f8>> accessed 24 June 2025. See also Michael Fischer, ‘Research: The largest corporate fines in 2024 so far’ (*Tradingpedia* 14 May 2024), <<https://www.tradingpedia.com/2024/05/02/research-the-largest-corporate-fines-in-2024-so-far/>> accessed 24 June 2025, which compares the fines imposed on some of the biggest tech companies with their annual revenue.
- 123 AIA, arts 73(11) and 73(8) combined with Market Surveillance Regulation (n 118), arts 19(1) and 20 and Regulation (EU) 2023/988 of the European Parliament and of the Council of 10 May 2023 on general product safety (Product Safety Regulation) [2023] OJ L135/1, art 26.
- 124 AIA, art 73(7).

- 125 AIA, art 77(1).
- 126 The Board is composed of one representative per MS, while the AI Office and the European Data Protection Supervisor participate without vote. Its main task is to advise and assist the Commission and the Member States in order to facilitate the consistent and effective application of the AIA—see AIA, arts 65–66.
- 127 AIA, art 57(16).
- 128 above n 14.
- 129 See Whistleblower Directive, arts 1 and 4.
- 130 See Whistleblower Directive, arts 7–14.
- 131 *ibid.*
- 132 *ibid*, arts 19–22.
- 133 *ibid*, arts 16–17.
- 134 *ibid*, art 18.
- 135 See Whistleblower Directive, art 5(1).
- 136 See Interim Measures for the Management of Generative Artificial Intelligence Services (2023), arts 14, 15, and 18, unofficial translation in English at <<https://www.chinalawtranslate.com/en/generative-ai-interim/>>, accessed 24 June 2025); Provisions on the Administration of Deep Synthesis Internet Information Services (2022), arts 10–12, unofficial translation in English at <<https://www.chinalawtranslate.com/en/deep-synthesis/>>; Provisions on the Management of Algorithmic Recommendations in Internet Information Services (2021), arts 9 and 30, unofficial translation in English at <<https://www.chinalawtranslate.com/en/algorithms/>>, accessed 24 June 2025. It is interesting that Chinese law obliges service providers to have also user complaint mechanisms in place.
- 137 See Basic Law on Artificial Intelligence Development, art 32(2), <<http://likms.assembly.go.kr/law/lawsLawtInqyDetl1010.do>>, accessed 24 June 2025 (in Korean, translated through machine translation).
- 138 See e.g. Gian Volpicelli and others, ‘Trump Administration Pressures Europe

to Ditch AI Rulebook' (*Bloomberg*, 25 April 2025)

<<https://www.bloomberg.com/news/articles/2025-04-25/trump-administration-presents-europe-to-reject-ai-rulebook>> accessed 24 June 2025.

- 139 According to the International Civil Aviation Organisation (ICAO), the global accident rate for 2023 was 1.87 accidents per million flight departures, for 2022 the respective rate was 2.05, and for 2021 it was 1.93. See ICAO, 'Safety Report, 2024 edition' <https://www.icao.int/sites/default/files/sp-files/safety/Documents/ICAO_SR_2024.pdf> accessed 24 June 2025, 12.
- 140 Laurin Weissinger, 'AI, Complexity and Regulation' (14 October 2021) <<https://ssrn.com/abstract=3943968>> 2.
- 141 Weissinger (n 140) 7; Brian Judge and others, 'When code isn't law: rethinking regulation for artificial intelligence', (2025) 44(1) *Policy and Society* 85, 90.
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- 143 One prominent, recent example are the accidents of two Boeing 737 Max, the Lion Air Flight 610 (JT610) and Ethiopian Airlines Flight 302 (ET302), in which the automated stabilization system of the aircraft (MCAS) malfunctioned during take-off and the pilots had a very short time margin seconds to recognize the problem and react, while being unaware of the exact function of the system and thus unable to predict its behaviour—see details from a human factors perspective in Immanuel Barshi et al, 'Analyses of the Boeing 737-MAX Accidents: Formal Models and Psychological Perspectives' <<https://ntrs.nasa.gov/citations/20230006844>> accessed 24 June 2025.
- 144 Dana Hall and Craig Trudell, 'A Fatal Tesla Crash Shows the Limits of Full Self-Driving' (*Bloomberg* 4 June 2025)

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<<https://www.nytimes.com/2024/10/23/technology/characterai-lawsuit-teen-suicide.html>> accessed 24 June 2025; Imane El Attilah, 'Man Ends his Life after an AI chatbot 'encouraged' him to Sacrifice Himself to Stop Climate Change' (*Euronews* 31 March 2023)
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- 150 See, for example, on the economic impact of grounding of the fleet of Boeing 737 Max following two deadly accidents Ted Reed, 'New Report Puts Impact Of Boeing 737 MAX Grounding at \$4.1 Billion' (*Forbes* 10 August 2019)
<<https://www.forbes.com/sites/tedreed/2019/08/10/new-report-puts-impact-of-boeing-737-max-grounding-at-41-billion/>> accessed 24 June 2025.
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<<https://www.cnbc.com/2024/01/30/boeings-reputation-is-taking-a-hit-survey-shows.html>> accessed 24 June 2025.
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<[https://economics.mit.edu/sites/default/files/publications/Harms%20of%](https://economics.mit.edu/sites/default/files/publications/Harms%20of%20AI.pdf)

- [20AI.pdf](#)>; NIST (n 64), paras 2.1-2.12; Dan Hendrycks and others, 'An Overview of Catastrophic AI Risks', 9 October 2023
<<https://arxiv.org/abs/2306.12001v6>> accessed 24 June 2025.
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- 154 *ibid.*
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<<https://www.datasciencecentral.com/an-overview-of-the-role-data-plays-in-ai-development/>> accessed 24 June 2025 and more details in Antonio V Silva Neto and others, 'Safety Assurance of Artificial Intelligence-Based Systems: A Systematic Literature Review on the State of the Art and Guidelines for Future Work' (2022) 10 *IEEE Access* 130733–130770.
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- 165 *ibid.*, art 54(l).
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- 167 Milde (n 166) 175, 179, who observes that the force of the SARPs could be compared with that of the ‘law of gravity’.
- 168 Consolidated version of the Treaty on the Functioning of the European Union [2016] OJ C202/1, art 4(2)(g).
- 169 See details in ICAO, *Annex 13 Aircraft Accident and Incident Investigation* (13th ed., July 2024), Chapter 1 Definitions; ICAO, *Annex 19 Safety Management Systems* (2nd ed., July 2016), Chapter 1 Definitions.
- 170 *ibid.*
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- 172 ICAO, *Annex 13* (n 169), Chapter 1 Definitions, Note 1 to ‘serious incident’.

- 173 Regulation (EU) No 376/2014 of the European Parliament and of the Council of 3 April 2014 on the reporting, analysis and follow-up of occurrences in civil aviation [2014] OJ L122/18, art 2(6).
- 174 Annex 19 (n 169), Standard 5.1.1. See also Annex 13, Standards 8.1 and 8.3. There is a distinction between *safety data*, which are the facts and sets of safety values collected from various sources, and *safety information*, which is the product of data processing, organization or analysis. See Annex 19, Chapter 1 Definitions.
- 175 Annex 19 (n 169), Standard 5.2.1.
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- 180 See Regulation 376/2014, art 4(1). The predefined categories concern (i) the operation of the aircraft, (ii) technical conditions, maintenance and repair of the aircraft, (iii) air navigation services and facilities, and (iv) aerodromes and ground services.
- 181 *ibid*, art 4(2) and (3).
- 182 *ibid*, art 4(4).
- 183 *ibid*, art 4(6). Such persons are e.g. the pilot-in-command of an aircraft, persons engaged in designing, manufacturing, continuous airworthiness monitoring, maintaining or modifying an aircraft, persons signing airworthiness review certificates etc.
- 184 *ibid*, art 5(1)–(4); ICAO Annex 19 (n 169), Standard 5.1.3.
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- 195 *ibid*, recital (10).
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- 206 *ibid*, C2–C5.
- 207 Regulation 376/2014, art 8(1).
- 208 *ibid*, art 9(1).
- 209 *ibid*, art 13(9).
- 210 See Francesca Pellegrino, *The just culture principles in aviation law* (Springer 2019) 12-16.
- 211 Regulation 376/2014, art 2(12).
- 212 Eurocontrol, 'Systems Thinking for Safety: Ten Principles, A White Paper' (August 2014) <<https://skybrary.aero/sites/default/files/bookshelf/2882.pdf>> accessed 24 June 2025, 12.
- 213 *ibid*.
- 214 Regulation 376/2014, art 15(2) and art 16; ICAO, *Annex 19* (n 169), Appendix 3, Standard 2.1
- 215 *ibid*, art 15(1)-(2); ICAO Annex 19 (n 169), Standard 5.3.
- 216 Pellegrino (n 210), 47; Francis Schubert, 'Legal Barriers to a Safety Culture in Aviation' (2004) 29 *Annals Air & Space L* 19, 64.
- 217 Regulation 376/2014, art 16 (6)-(7) and (9)-(10). See also the definition of 'just culture' mentioned above and ICAO Annex 19 (n 169), Appendix 3, Standard 3.
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- 234 Regulation (EU) 2018/1725 of the European Parliament and of the Council of 23 October 2018 on the protection of natural persons with regard to the processing of personal data by the Union institutions, bodies, offices and agencies and on the free movement of such data [2018] OJ L295/39.
- 235 Regulation 396/2014, art 20.
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242 Regulation 396/2014, art 9(1).

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244 Regulation 396/2014, art. 14 and recital (16). On the ADREP taxonomy see <<https://www2023.icao.int/safety/airnavigation/AIG/Pages/ADREP-Taxonomies.aspx>> accessed 11 February 2026.

245 *ibid*, art 10(1).

246 See *ibid*, Annex II.

247 *ibid*, art 10(2)-(4) and Annex II. For security reasons, direct access is not granted in such cases.

248 See Regulation 376/2014, art 11.

249 *ibid*, art 11(4).

250 *ibid*, art 12.

251 *ibid*, art 13(3).

252 Sebastian Sieberichs and Annette Kluge, 'Why Commercial Pilots Voluntarily Report Self-inflicted Incidents' (2021) 11 *Aviation Psychology and Applied Human Factors* 98, 103; Robert Baron, 'Feedback is Key to a Successful Safety Reporting System' (*LinkedIn*, 16 August 2022) <<https://www.linkedin.com/pulse/feedback-key-successful-safety-reporting-system-baron-ph-d>> accessed 24 June 2025.

253 Regulation 376/2014, art 13 (11).

254 *ibid*, art 14(4).

255 Regulation (EU) 2018/1139 of the European Parliament and of the Council of 4 July 2018 on common rules in the field of civil aviation and establishing a European Union Aviation Safety Agency [2018] OJ L 212/1, art 72(7).

256 See CM Holloway and CW Johnson, 'A Look at Aircraft Accident Analysis in the Early Days: Do Early 20th Century Accident Investigation Techniques Have Any Lessons for Today?' in *2007 2nd Institution of Engineering and*

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June 2025, 235-240; US Federal Aviation Administration, 'History of Aviation Safety Oversight in the United States', DOT/FAA/AR-08/39 (July 2008)

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- 257 Article 26 provides: 'In the event of an accident to an aircraft of a contracting State occurring in the territory of another contracting State, and involving death or serious injury, or indicating serious technical defect in the aircraft or air navigation facilities, the State in which the accident occurs will institute an inquiry into the circumstances of the accident, in accordance, so far as its laws permit, with the procedure which may be recommended by the International Civil Aviation Organization. The State in which the aircraft is registered shall be given the opportunity to appoint observers to be present at the inquiry and the State holding the inquiry shall communicate the report and findings in the matter to that State.'
- 258 See ICAO, *Annex 13 Aircraft Accident and Incident Investigation* (12th edn, 2020), p. ix.
- 259 Regulation (EU) No 996/2010 of the European Parliament and of the Council of 20 October 2010 on the investigation and prevention of accidents and incidents in civil aviation [2010] OJ L295/35.
- 260 ICAO, *Annex 13*, p 1-2.
- 261 ICAO, *Annex 19 Safety Management* (1st edn, 2013), Standard 1.3 and Attachment A, para 1.3.
- 262 'Safety occurrences investigation', SKYBRARY, <<https://skybrary.aero/articles/safety-occurrence-investigation>> accessed 24 June 2025.
- 263 Regulation 996/2010, art 5(1); ICAO, *Annex 13* (n 169), Standard 5.1, provides also for the possibility of delegation to a specialized regional aviation safety organization.

- 264 Regulation 996/2010, art 6(2), which, however, does not provide for the possibility of delegation to an aviation safety organization; ICAO Annex 13 (n 169), Standard 5.1. An example of such delegation was the investigation conducted by the Dutch Safety Board into the downing of Malaysian Airlines Flight MH17 on 17 July 2014 over the region of Donetsk in Eastern Ukraine—see Dutch Safety Board, ‘Crash MH17, 17 July 2014’
<<https://onderzoeksraad.nl/en/onderzoek/crash-mh17-17-july-2014/>> accessed 24 June 2025.
- 265 UK Air Accidents Investigation Branch, ‘Annual safety review 2023’ (28 March 2024)
<https://assets.publishing.service.gov.uk/media/665ddb537b792ffff71a866f/AAIB_Annual_Safety_Review_2023.pdf> accessed 24 June 2025, 8.
- 266 Regulation 996/2010, art 10(1); ICAO, *Annex 13* (n 169), Standard 5.18 and 5.23.
- 267 See Regulation 996/2010, art 21(3)-(5); ICAO, *Annex 13* (n 169), Standard 5.27.
- 268 Regulation 996/2010, art 8.
- 269 See *ibid*, arts 8 and 10; ICAO, *Annex 13* (n 169), Standards 5.23-5.27.
- 270 See Regulation 996/2010, art 7.
- 271 *ibid*, arts 1(1) and 5(6); ICAO, *Annex 13* (n 169), Standard 3.1.
- 272 Regulation 996/2010, art 4(1)-(3); ICAO, *Annex 13* (n 169), Standard 3.2.
- 273 Regulation 996/2010, art 4(6).
- 274 Schubert (n 216) 33.
- 275 Smart (n 161) 112; Paul Stephen Dempsey, ‘Independence of Aviation Safety Investigation Authorities: Keeping the Foxes from the Henhouse’ (2010) 75 J AIR L & COM 223, 282–283.
- 276 ICAO, *Annex 13*, Definition of ‘investigation’, p 1-2.
- 277 Regulation 996/2010, art 5(3).
- 278 ‘Safety occurrences investigation’, SKYBRARY
<<https://skybrary.aero/articles/safety-occurrence-investigation>> accessed

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- 279 See *ibid* for a summary description of the investigation phases. See also ICAO, *Manual of Aircraft Accident and Incident Investigation – Part III Investigations*, Doc 9756 (1st ed, 2012), para 1.1., which distinguishes three phases: collection of data, Analysis of data and Presentation of data.
- 280 Regulation 996/2010, art 11; ICAO, *Annex 13*, Standards 5.5–5.6.
- 281 Regulation 996/2010, art 13; ICAO, *Annex 13* (n 169), Standard 3.3.
- 282 Regulation 996/2010, art 12; ICAO, *Annex 13* (n 169), Standards 5.4 and 5.10.
- 283 Regulation 996/2010, art 12(1), last sentence; ICAO, *Annex 13*, Standard 5.4.3.
- 284 Regulation 996/2010, arts 14–15; ICAO, *Annex 13* (n 169), Standards 5.12.2, 5.12.3, 5.12.5, and 5.12.6.
- 285 Aaron S Dietz and others, ‘Constructing Accurate and Precise Timelines for Major Aviation Accident Investigations’, 2009 International Symposium on Aviation Psychology,
<https://corescholar.libraries.wright.edu/isap_2009/91> accessed 24 June 2025, 142, 143.
- 286 Kym Bills and others, ‘Major aviation accident investigation methodologies used by ITSA members’, (2023) 168 *Safety Science* 106315.
- 287 ICAO, *Annex 13* (n 169), Standards 7.1–7.4.
- 288 Regulation 996/2010 art 16(3), second sentence; ICAO, *Annex 13* (n 169), Standard 6.2.
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- 290 Regulation 996/2010, art 16(6); ICAO, *Annex 13* (n 169), Standard 6.5.
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- 292 *ibid*.
- 293 *ibid*, Definitions.
- 294 Smart (n 161), 112.

- 295 Regulation 996/2010, art 16(1), third sentence; ICAO Annex 13 (n 169), Appendix I.
- 296 Regulation 996/2010, art 17(3).
- 297 *ibid*, art 17(1); ICAO Annex 13 (n 169), Standards 6.8 and 6.8.1.
- 298 ICAO Annex 13 (n 169), Note to Standard 6.8.
- 299 *ibid*, p 1-3, definition of SGRC.
- 300 *ibid*, Standard 6.9.1. See also the relevant section of the ICAO website <[https://www.icao.int/safety/airnavigation/AIG/Pages/Safety-Recommendations-of-Global-Concern-\(SRGC\).aspx](https://www.icao.int/safety/airnavigation/AIG/Pages/Safety-Recommendations-of-Global-Concern-(SRGC).aspx)> accessed 24 June 2025.
- 301 Regulation 996/2010, art 18(1); ICAO, *Annex 13* (n 169), Standard 6.10.
- 302 Regulation 996/2010, art 18(2)-(3); ICAO, *Annex 13* (n 169), Standard 6.11.
- 303 Regulation 996/2010, art 18(4)-(5); ICAO, *Annex 13* (n 169), Standard 6.11–6.12.
- 304 See e.g. Clinton V Oster and others, ‘Analyzing aviation safety: Problems, challenges, opportunities’ (2013) 43 *Research in Transportation Economics* 148, 161–163.
- 305 See e.g. ‘US NTSB Chairman stresses data and cooperation as keys I the future of accident investigation’, (2012) 67(1) *ICAO Journal* 14, 15.
- 306 Cf. Jeanne Casperson and others, ‘The Social Amplification of Risk: Assessing Fifteen Years of Research and Theory’ in Paul Slovic (ed.) *The Feeling of Risk* (Earthscan 2010) 317–344.
- 307 E.g. the German air carrier Germanwings had to cease operations and be rebranded to Eurowings following the crash of flight 9525 on 24 March 2015, in which all 150 persons on board were killed—see ‘One year since the Germanwings disaster’ (*Deutsche Welle*, 24 March 2016) <<https://www.dw.com/en/one-year-on-victims-families-gather-to-mark-germanwings-crash/a-19137553>> accessed 24 June 2025. The Cypriot air carrier Helios Airways was rebranded to Ajet after the fatal accident of 14 August 2005, in which all 121 occupants were killed, and a few months later ceased operations—see Bill Giannopoulos ‘Helios Airways, inspired by the

radiance of the Greek sun god Apollo, emerged as a groundbreaking force in Cypriot aviation' (*Greek City Times* 6 May 2024)

<<https://greekcitytimes.com/2024/05/06/helios-airways-inspired-by-the-radiance-of-the-greek-sun-god-apollo-emerged-as-a-groundbreaking-force-in-cypriot-aviation/>> accessed 24 June 2025.

308 George Leloudas, *Risk and liability in air law* (Informa Law 2009), paras 3.22–3.29, 3.43–3.56.

309 See e.g. João da Silva, 'Criminal Charges Recommended against Boeing', (*BBC* 25 June 2024) <<https://www.bbc.com/news/articles/c0666zr4010o>> accessed 24 June 2025.

310 Will Hunt, 'The Flight to Safety-critical AI. Lessons in AI Safety from the Aviation Industry', UC Berkeley Center for Long-Term Cybersecurity White Paper (August 2020) <<https://cltc.berkeley.edu/publication/new-report-the-flight-to-safety-critical-ai-lessons-in-ai-safety-from-the-aviation-industry/>> accessed 24 June 2025, 18. See also Dempsey (n 275) 280, who notes that the results of an accident investigation can be politically and economically devastating.

311 Compare AIA, Annex III. See also Natalie Smuha, 'Beyond the Individual: Governing AI's Societal Harm' (2021) 10 *Internet Policy Review*, <<https://doi.org/10.14763/2021.3.1574>> accessed 24 June 2025, 6–12.

312 See I Glenn Cohen and others, 'Navigating the New Risks and Regulatory Challenges of GenAI' (*Harvard Business Review*, 20 November 2023) <<https://hbr.org/2023/11/navigating-the-new-risks-and-regulatory-challenges-of-genai>> accessed 24 June 2025.

313 Claudia Wilson, 'AI Safety and the US-China Arms Race' (*Center for AI Policy*, 29 October 2024) <<https://www.centeraipolicy.org/work/ai-safety-and-the-us-china-arms-race>> accessed 24 June 2025; Haileleol Tibebu, 'DeepSeek and the Race to AGI: How Global AI Competition Puts Ethical Accountability at Risk' (*Tech Policy Press*, 29 January 2025) <<https://www.techpolicy.press/deepseek-and-the-race-to-agi-how-global-ai-competition-puts-ethical-accountability-at-risk/>> accessed 24 June 2025.

314 Matthijs Maas, 'Regulating for "Normal AI Accidents" – Operational Lessons for the Responsible Governance of AI Deployment' in Association for Computing Machinery, *Proceedings of the 2018 AAAI/ACM Conference on AI*,

Ethics, and Society (AIES '18) (New York 2018), 223, 225–226. See also Oliver Guest and others, ‘Safeguarding the Safeguards’, (*Institute for AI Policy and Strategy*, 14 December 2023), <<https://www.iaps.ai/research/safeguarding-the-safeguards>> accessed 24 June 2025, 22, referring to the risk of ‘safety washing’, i.e. AI developers appearing to improve safety without actually reducing risks.

- 315 Saurabh Bagchi, ‘What is a Black Box? A Computer Scientist Explains What it Means When the Inner Workings of AIs are Hidden’, (*The Conversation* 22 May 2023) <<https://theconversation.com/what-is-a-black-box-a-computer-scientist-explains-what-it-means-when-the-inner-workings-of-ais-are-hidden-203888>> accessed 24 June 2025.
- 316 Noam Kolt and others, ‘Responsible Reporting for Frontier AI Development’ in Association for Computing Machinery, *Proceedings of the Seventh AAAI/ACM Conference on AI, Ethics, and Society (AIES 2024)* <<https://ojs.aaai.org/index.php/AIES/article/download/31678/33845/35742>> accessed 24 June 2025, 768, 773. See also AIA, arts 78 and 74(13).
- 317 See for the US at federal level Executive Office of the President of the US, ‘America’s AI Action Plan’ (July 2025) <<https://www.whitehouse.gov/wp-content/uploads/2025/07/Americas-AI-Action-Plan.pdf>> accessed 28 July 2025, 3.
- 318 ICAO, *Safety Management Manual* (n 176), para 3.1.
- 319 See Ben Shneiderman, ‘Bridging the Gap between Ethics and Practice: Guidelines for Reliable, Safe, and Trustworthy Human-centered AI Systems’ (2020) 10 ACM Trans Interact Intell Syst, <<https://doi.org/10.1145/3419764>>, 26:10–26:16, who identifies five main themes in this regard: (i) Leadership commitment to safety, (ii) hiring and training oriented to safety, (iii) extensive reporting of failures and near misses, (iv) internal review boards for problems and future plans, and (v) alignment with industry standards and accepted best practices.
- 320 See ICAO, *Safety Management Manual* (n 176), para 3.2.5.1, which underlines the need for regulators or organization managers to build trust with the organizations or personnel who are reporting the information.
- 321 cf. Regulation 996/2010, art 2(12).

- 322 Recital (j): ‘The Signatories recognise that the reporting of a serious incident is not an admission of wrongdoing’.
- 323 Measure 3.5 Post Market Monitoring, which mentions as examples of post-market monitoring methods the collection of end-user feedback, the provision of (anonymous) reporting channels, and the provision of serious incidents reporting forms, combined with Measure 9.1 Measures for Serious Incident Identification, which refers to such methods.
- 324 cf. ICAO, *Safety Management Manual* (n 176), para 5.2.5.5, which provides that two main aspects should be considered when deciding whether an incident should be classified as a serious incident: (a) if there were circumstances indicating a high probability of an accident and (b) if the accident was avoided only due to providence.
- 325 NASA (n 190) 3; Violet Turri and Rachel Dzombak, ‘Why We Need to Know More: Exploring the State of AI Incident Documentation Practices’ in *Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society (AIES ’23)* (Association for Computing Machinery 2023) <<https://dl.acm.org/doi/10.1145/3600211.3604700>> accessed 11 February 2026, 576, 582.
- 326 Kolt and others (n 316), 773–774; Miles Brundage and others, ‘Towards Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims’ (2020) <<https://arxiv.org/abs/2004.07213>> accessed 24 June 2025, 19.
- 327 See also Kolt and others (n 316), 768, who note that there is growing consensus among experts in AI safety and governance that reporting safety information to trusted actors in government and industry is key to responding effectively to the new and emerging risks presented by frontier AI systems.
- 328 Under Whistleblower Directive, art 26, its transposition deadline expired on 17 December 2021 and all MS have transposed it, although challenges remain—see Report from the Commission to the European Parliament and the Council on the implementation and application of Directive (EU) 2019/1937 of the European Parliament and of the Council of 23 October 2019 on the protection of persons who report breaches of Union law, COM (2024) 269 final.
- 329 See also OECD, ‘Stocktaking for the Development of an AI Incident

Definition' OECD Artificial Intelligence Papers, No. 4 (OECD Publishing 2023) <<https://doi.org/10.1787/c323ac71-en>> accessed 24 June 2025, Annex A, which includes issues of incentives to report (*why*) in the common incident-reporting framework.

- 330 ICAO, *Safety Management Manual* (n 176), para 9.4.4.9.
- 331 Feiler and König (n 45) para 3.49.
- 332 OECD, 'Defining AI Incidents and Related Terms' (n 43), 11–14.
- 333 See *ibid*, 12, which underlines that assessing the 'seriousness' of an AI incident is highly context-dependent, and each State may define it differently.
- 334 Ren Bin Lee Dixon and Heather Frase, 'AI Incidents: Key Components for a Mandatory Reporting Regime' (Center for Security and Emerging Technology, January 2025) <<https://cset.georgetown.edu/publication/ai-incidents-key-components-for-a-mandatory-reporting-regime/>> accessed 24 June 2025, 17.
- 335 Compare ICAO, *Safety Management Manual* (n 176), para 5.2.5.1, which clarifies that mandatory safety reporting systems should aim to capture all valuable information about an occurrence, including what happened, where, when, and to whom the report is addressed, alongside some specific hazards known to contribute to accidents, such as meteorological conditions, volcanic activity, etc.
- 336 AIA, art 73(7).
- 337 OECD, 'Towards a Common Reporting Framework for AI Incidents', OECD Artificial Intelligence Papers, No. 34 (OECD Publishing 2025) <<https://doi.org/10.1787/f326d4ac-en>> accessed 24 June 2025, Annex B; Dixon and Frase (n 334) 7-16.
- 338 AIA, art 9.
- 339 AIA, art 72.
- 340 AIA, art 1(1) and recital (1).
- 341 See Regulation 376/2014, arts 6(2) and 4(5) second sentence; Commission

Implementing Regulation (EU) 2015/1018 of 29 June 2015 laying down a list classifying occurrences in civil aviation to be mandatorily reported according to Regulation (EU) No 376/2014 of the European Parliament and of the Council [2015] OJ L163/1, Annex V.

- 342 Brundage and others (n 326), 20. Compare Laura Kabelka, ‘EU’s Cyber Incident Reporting Mechanism Does Not Work, Agency Chief warns’, (*Euractiv*, 27 April 2022) <<https://www.euractiv.com/section/cybersecurity/news/eus-cyber-incident-reporting-mechanism-does-not-work-agency-chief-warns/>> accessed 24 June 2025.
- 343 Sieberichs and Kluge (n 252), 104–105.
- 344 AIA, art 74(1) combined with Regulation 2019/1020, art 11(1).
- 345 Dixon and Frase (n 334) 17.
- 346 Apart from the rules in aviation, independent investigations are provided for in Directive 2009/18/EC of the European Parliament and of the Council of 23 April 2009 establishing the fundamental principles governing the investigation of accidents in the maritime transport sector [2009] OJ L131/114, art 8(1) and in Directive (EU) 2016/798 of the European Parliament and of the Council of 11 May 2016 on railway safety [2016] OJ L138/102, art 22(1).
- 347 See Martin Culter, ‘AI Talent War Heats Up in Europe’ (*Reuters*, 11 March 2024) <<https://www.reuters.com/technology/ai-talent-war-heats-up-europe-2024-03-11/>> accessed 24 June 2025; ‘The war for talent is heating up’ (*The Economist*, 8 June 2024) <<https://www.economist.com/business/2024/06/08/the-war-for-ai-talent-is-heating-up>>, accessed 24 June 2025.
- 348 See David Coen and Colin Provost, ‘Revolving Doors’ in Phil Harris and others (eds) *The Palgrave Encyclopedia of Interest Groups, Lobbying and Public Affairs* (Palgrave Macmillan, 2022) 1170, 1170–1172.
- 349 E.g. Medical Devices Regulation (n 11), art 87; GDPR (n 12), art 33; Directive (EU) 2022/2557 on railway safety (n 346), art 20; Directive (EU) 2022/2555 of the European Parliament and of the Council of 14 December 2022 on measures for a high common level of cybersecurity across the Union,

amending Regulation (EU) No 910/2014 and Directive (EU) 2018/1972, and repealing Directive (EU) 2016/1148 (NIS 2 Directive) [2022] OJ L333/80, art 23.

350 AIA, arts 40, 95 and 96.

351 AIA, art 13(1) and (3), art 11(1) combined with Annex IV(2).

352 See High Level Expert Group on AI, 'Ethics Guidelines for Trustworthy AI' (2019) <<https://ec.europa.eu/futurium/en/ai-alliance-consultation.1.html>> accessed 24 June 2025, 14–15.

353 See Sabine Theis and others, 'Requirements for Explainability and Acceptance of Artificial Intelligence in Collaborative Work' in Helmut Degen and Stavroula Ntoa (eds) *Artificial Intelligence in HCI* (Springer 2023), 355, 370–371; Pascal Koenig, 'Attitudes Toward Artificial Intelligence: Combining Three Theoretical Perspectives on Technology Acceptance' (2025) 40 AI & Soc 13333, 1342–1343.

354 AIA, art 78(1)(a) and (2).

355 cf. Directive (EU) 2016/943 of the European Parliament and of the Council of 8 June 2016 on the protection of undisclosed know-how and business information (trade secrets) against their unlawful acquisition, use and disclosure [2016] OJ L 157/1, art 5 regarding exceptions to the protection of trade secrets; CJEU C-588/21 P *Public.Resource.Org and Right to Know v Commission and Others*, ECLI:EU:C:2024:201, paras 23–31 and 81–86 regarding public access to European harmonised standards, allegedly protected by copyright.

356 AIA, art 68.

357 AIA, art 69.

358 See Mia Hunt, 'Beyond Pay and Promotion: How Governments can Attract and Retain Staff in a Fast-changing World' (*Global Government Forum*, 14 February 2023) <<https://www.globalgovernmentforum.com/beyond-pay-and-promotion-how-governments-can-attract-and-retain-staff-in-a-fast-changing-world/>> accessed 24 June 2025; Amrita Datar and others, 'Government can win the talent race—Here's how' (*Deloitte*, 23 May 2022)

<<https://www2.deloitte.com/us/en/insights/industry/public-sector/talent-war-government.html>> accessed 24 June 2025.

- 359 See Regulation 996/2010, art 6(2).
- 360 David Coen and Colin Provost (n 348), 1173–1174; Silvia Kotanidis, ‘Rules on ‘revolving doors’ in the EU’ (European Parliamentary Research Service, 2024) <<https://tinyurl.com/3fjdc8cn>>, accessed 11 February 2026, 1–3.
- 361 Measure 9.1 Methods for serious incident identification.
- 362 See High Level Expert Group on AI (n 352), 19–20.
- 363 Turri and Dzombak (n 325) 581.
- 364 Giampiero Lupo, ‘Risky Artificial Intelligence: The Role of Incidents in the Path to AI Regulation’ (2023) *Law, Technology and Humans* 5, 133, 144–147; Will Hunt, ‘The Flight to Safety-critical AI’ (2020) UC Berkeley Center for Long-Term Cybersecurity <<https://cltc.berkeley.edu/publication/new-report-the-flight-to-safety-critical-ai-lessons-in-ai-safety-from-the-aviation-industry/>> accessed 24 June 2025, 25.
- 365 AIA, art 68.
- 366 See High Level Expert Group on AI (n 352), 16–18.
- 367 Compare ICAO, *Safety Management Manual* (n 176), para 7.2.2: ‘Aviation safety is not the sole responsibility of States or service providers. It is a shared responsibility to which all stakeholders should contribute by, among other things, providing relevant data and information through safety reports.’

Author notes

Conflicts of interest. None declared.

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